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Processing-bias correction with DEBIAS-M improves cross-study generalization of microbiome-based prediction models

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Abstract

Every step in common microbiome profiling protocols has variable efficiency for each microbe, e.g., different DNA extraction efficiency for Gram-positive bacteria. These processing biases impede the identification of signals that are biologically interpretable and generalizable across studies. “Batch-correction” methods have been used to address these issues computationally with some success, but they are largely non-interpretable and often require the use of an outcome variable in a manner which risks overfitting. We present DEBIAS-M (**D**omain adaptation with **E**stimation and **B**atch **I**ntegration **A**cross **S**tudies of the **M**icrobiome), an interpretable framework for inference and correction of processing bias, which facilitates domain adaptation in microbiome studies. DEBIAS-M learns bias-correction factors for each microbe in each batch that

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Author contributions

G.I.A. and T.K. conceived and designed the study, designed analyses, interpreted the results, and wrote the manuscript with input from all authors. G.I.A. conceived and wrote DEBIAS-M and conducted all analyses. S.E., J.B., A.B.K and H.P. acquired, harmonized, and processed data as well as assisted with analyses. A.-C.U. and I.P. provided critical comments on the design, method, and analysis. T.K. supervised the study.

Code availability

DEBIAS-M is available from <https://korem-lab.github.io/DEBIAS-M/>. Code used to generate all analyses and plots can be found at <https://github.com/korem-lab/DEBIAS-M-Analysis/>.

Competing interests

The authors declare no competing interests.

simultaneously minimize batch effects and maximize cross-study associations with phenotypes. Using diverse benchmarks including 16S and metagenomic sequencing, classification and regression, and a variety of clinical and molecular targets, we demonstrate that using DEBIAS-M improves cross-study prediction accuracy compared to commonly used batch-correction methods. Notably, we show that the inferred bias-correction factors are stable, interpretable, and strongly associated with specific experimental protocols. Overall, we show that DEBIAS-M facilitates improved modeling of microbiome data and identification of interpretable signals that generalize across studies.

Introduction

A hallmark of a robust scientific analysis is that its conclusions generalize beyond a specific protocol, study, or population. Such generalization offers strong evidence that the findings are not the result of the particularities of one experiment, reduces the impact of confounding variables, and, in general, lowers the risk for spurious findings. For prediction models, external and independent validation is imperative for assessing the generalizability of the model¹. The ability to train generalizable models across datasets also provides increased sample size in settings where data from many smaller studies is available², such as studies of the vaginal microbiome in preterm birth³, or the gut microbiome in colorectal cancer^{4,5}.

In profiling microbiomes, challenges in generalizability stem not only from biological variability (e.g., differences between populations or medical practices), but also from technical variability between experimental protocols, facilities, and bioinformatic pipelines. Such variability has been noted repeatedly^{6–12}, and affects the replicability and interpretability of microbiome data¹³. While this has prompted calls for field-wide protocol standardization, others doubt the utility of this strategy¹⁴, and, in practice, it has not been widely adopted.

In a landmark study, McLaren, Willis & Callahan provided and validated a mathematical framework for consideration of variability in microbiome studies¹⁵, drawing in part on previous efforts¹⁰. Under this framework, technical variability manifests as a taxon-specific multiplicative bias that stems from differential efficiencies of different experimental and analytic processes, including DNA extraction, PCR, sequencing, and bioinformatics analyses. As each taxon and study have their own unique biases, associations (e.g., with a phenotype) measured in one study might not replicate in a second one (Fig. 1a). Importantly, McLaren et al. show that even if identical taxon-specific biases apply across samples, their effect on each sample depends on its microbial composition, and is therefore not eliminated even with rigorous standardization of experimental factors¹⁵. Hence, there is a pressing need for methods that can retrospectively infer and correct processing biases introduced during microbiome profiling.

Many approaches relevant to this challenge have been developed under the general term of batch-correction methodologies, some for microbiome data^{16–20} and others adopted from other fields^{21–24}. While these methods show benefit in correcting batch effects, many of them are suitable only for association testing, rather than predictive modeling. Other methods which have been used for predictive modeling require the use of the outcome

variable, which risks overfitting and limits the ability to evaluate generalization on held-out data. Lastly and importantly, the changes these methods make to the data are largely non-interpretable²⁵, as recently highlighted with respect to an application of voom-SNM^{22,23} to a tumor-microbiome analysis²⁶, in which values were introduced to features that were initially very sparse²⁷.

Here, we present “**Domain adaptation with phenotype Estimation and Batch Integration Across Studies of the Microbiome**” (DEBIAS-M), a method for inference and correction of microbiome processing bias that is designed to operate in the context of multiple batches or studies. DEBIAS-M infers taxon- and batch-specific processing biases by finding batch-wise patterns, which, when corrected for, both reduce differences between batches and improve the overall association with phenotypes of interest. We demonstrate that DEBIAS-M outperforms batch-correction methods using diverse benchmarks of HIV, colorectal cancer, cervical neoplasia, and metabolite levels predictions across gut and vaginal microbiome studies, including the use of both metagenomics and 16S sequencing data. We further show that the bias-correction factors learned by DEBIAS-M are interpretable and strongly associated with experimental protocols. Finally, we demonstrate that incorporating DEBIAS-M into typical machine-learning pipelines improves prediction accuracy on held-out studies. Overall, DEBIAS-M offers the capability to leverage more datasets for explainable, stronger, and more replicable microbiome analyses.

Results

Description of DEBIAS-M

DEBIAS-M is a method for *in silico* detection and correction of processing bias. It takes as input read counts or relative abundances of taxa in microbiome samples from multiple studies or batches (hereafter collectively termed “batches”), and learns one multiplicative coefficient¹⁵ for each taxon in every batch, which corrects for the processing bias of that batch (Fig. 1a). Every sample is then renormalized before a downstream prediction model – identical across all batches – learns an association to a phenotype of interest (Fig. 1b; Methods). The bias and prediction parameters are optimized to maximize the prediction likelihood while minimizing the variation between batches. Samples for which the phenotype is unavailable or hidden, such as those with missing data or samples in a test set, are not included in the calculation of the prediction loss, and they are either only considered when minimizing cross-batch differences (Fig. 1b and Extended Data Fig. 1a–b) or not at all (“Online DEBIAS-M”, Extended Data Figs. 1d, 2). To facilitate broad utility, DEBIAS-M offers flexible modes of usage, including operations in log space, application for regression and classification, single- and multi-task learning, and online correction and prediction for previously unobserved batches (no retraining; Methods). DEBIAS-M is available as a python package from <https://korem-lab.github.io/DEBIAS-M/>, with a complete description available in the Methods.

DEBIAS-M outperforms alternatives in diverse benchmarks

We first sought to evaluate DEBIAS-M and compare it to popular batch-correction approaches, focusing on six: ComBat²¹, ConQuR¹⁸, percnorm¹⁶, voom-SNM^{22,23,26},

MMUPHin¹⁹, and PLSDA-batch²⁰, in addition to using the raw, uncorrected data. To this end, we set up four cross-study prediction benchmarks in which we applied each batch-correction method followed by a classifier of a relevant outcome (Methods). We used a leave-one-study-out approach, where a model was trained on all studies except one, and evaluated on the held-out study (Extended Data Fig. 1a–b). The underlying assumption of this benchmark is that improved batch correction should result in higher classification accuracy on held-out studies. To focus this benchmark on batch correction rather than elaborate machine-learning pipelines, we used linear models. No outcome labels from the held-out study were provided during batch correction or model fitting.

Our first benchmark, analyzed previously¹⁸, used publicly available gut 16S rRNA sequencing data collected from 17 case-control studies to classify HIV diagnosis²⁸ (N of 13–233 subjects per study, total of 1,032; Methods; Supplementary Table 1). We found that DEBIAS-M outperformed raw data and data corrected by alternative methods (median [IQR] area under the receiver operating characteristic curve [auROC] of 0.7 [0.61–0.75] for DEBIAS-M vs. 0.5 [0.45–0.53] for raw and 0.55 [0.50–0.61] across all other methods; Fisher’s multiple comparison of one-sided DeLong tests $p < 0.01$ for all pairwise comparisons with alternative methods; Fig. 2a; Supplementary Table 2).

We then performed a similar cross-study evaluation of predicting colorectal cancer from publicly-available gut metagenomic sequencing data^{4,5,29–34} (N of 53–128 subjects per study, total of 886; Supplementary Table 3; Methods). Once again, DEBIAS-M outperformed alternative methods, with raw data yielding median [IQR] auROCs of 0.58 [0.57–0.65] and alternative methods yielding 0.61 [0.56–0.68] (across all methods), compared to 0.76 [0.69–0.78] for DEBIAS-M ($p < 0.01$ for all; Fig. 2b; Supplementary Table 2).

Next, we performed a similar benchmark using the cervicovaginal microbiome to predict cervical neoplasia, a challenging scenario with potential translational implications. We compiled and uniformly analyzed data from five independent studies^{35–39} (N of 29–82 subjects per study, total of 322; Methods; Supplementary Table 4), and performed a cross-study evaluation of classification of cervical intraepithelial neoplasia, a precancerous state conveying high risk for cervical cancer. While the linear models trained on raw data and data corrected by alternative methods yielded median [IQR] auROCs of 0.50 [0.42–0.51] and 0.51 [0.47–0.58] (across all methods), respectively, DEBIAS-M had cross-study auROCs of 0.65 [0.55–0.65] (Fisher’s multiple comparison of one-sided DeLong $p < 0.02$ for all pairwise comparisons with DEBIAS-M; Fig. 2c; Supplementary Table 2). Finally, we compiled data from five studies that collected cervicovaginal microbiome from patients with and without cervical carcinoma^{35,38,40–42} (N=26–77 individuals per study, total of 206; Methods, Supplementary Table 5). In a similar benchmark for classifying cervical carcinoma, DEBIAS-M was also significantly more accurate than all alternatives (median [IQR] auROC of 0.74 [0.70–0.85] vs. 0.53 [0.50–0.61] across all other methods and 0.58 [0.57–0.61] for raw data; $p < 0.02$ for all pairwise comparisons with DEBIAS-M; Fig. 2d; Supplementary Table 2). Overall, DEBIAS-M demonstrates robust performance improvement compared with batch-correction methods across four different benchmarks

totaling 2,340 samples from 35 studies, including both vaginal and gut microbiome, 16S and metagenomic sequencing.

Online learning with DEBIAS-M has similar performance

Multiple strategies have been used^{18,20,26,43} with respect to using features (i.e., microbiome data) and outcomes (i.e., phenotypes) during batch correction, which impacts the evaluation of machine-learning models (Extended Data Fig. 1). Above we have used a common approach, in which features are available for all samples (from both the training and test sets) during batch correction, but the phenotype labels are only available for the training set (Extended Data Fig. 1a–b), ensuring no “information leakage” from the test set⁴³. We note, however, that in some previous studies batch-correction methods used phenotype labels from all samples during batch-correction, including the test set, thus allowing such leakage (Extended Data Fig. 1d), and that doing so in our benchmarks drastically inflates predictive performance (Extended Data Fig. 3a–b). Additionally, as concerns have been raised regarding the use of features and covariates from the test set during data processing⁴³, we developed a variation of DEBIAS-M that performs batch correction and model training using only the training set, and adapts the inferred correction factors to the test set only once the rest of the model parameters are frozen (Methods; Extended Data Fig. 1c). DEBIAS-M performs equivalently both when observing test-set features and when these remain “hidden” (Extended Data Figs. 1a,c, 2a–c), demonstrating that it is robust to training and evaluation strategy.

Log-additive DEBIAS-M successfully operates in log space

While many batch-correction methods operate in count or relative abundance space, the use of compositional transformations such as the centered log-ratio (clr) transformation is common. We therefore next checked how DEBIAS-M compares to other methods in this setting (Methods), using both a clr-transformation of standard DEBIAS-M output and a version of DEBIAS-M that performs log-additive correction (Methods). Log-additive DEBIAS-M outperformed the alternatives in all datasets also in this benchmark, demonstrating its robustness to operating in log space (Fisher’s multiple comparison of one-sided DeLong $p < 0.01$ for all but two pairwise comparisons; Extended Data Fig. 2d–g). However, a clr-transformation of the output of standard DEBIAS-M did not show a consistent improvement relative to other methods (Extended Data Fig. 2d–g), and in two cases was significantly worse than log-additive DEBIAS-M ($p < 0.01$; Extended Data Fig. 2e–f). We note that the additive property of biases in log-space means that they would have a weaker impact on associations, and is a potential reason for the oft-observed higher performance of microbiome-based models in log space.

DEBIAS-M is robust to dataset characteristics

To evaluate the performance of DEBIAS-M with respect to different dataset characteristics, we used two complementary benchmarks. First, we created synthetic simulations of multiple microbiome datasets under our generative model, applying the bias framework of McLaren et al.¹⁵ to data generated by a microbiome data simulator⁴⁴, and generating synthetic phenotype labels with varying association strength with the ground-truth (“pre-biased”) microbial abundances (Methods). We ran DEBIAS-M on this data, compared the Jensen-

Shannon divergence between the ground-truth pre-biased abundances and the output from DEBIAS-M on a held-out test batch (Fig. 3 and Extended Data Fig. 1a), and also evaluated the ability of DEBIAS-M to infer the ground-truth processing biases (Extended Data Fig. 4). As this evaluation is mostly under our generative model, we also created a second benchmark, in which we subsampled the colorectal cancer dataset (analyzed in Fig. 2b) and analyzed it similarly except we evaluated prediction accuracy (as ground-truth abundances or processing biases are not available; Methods; Extended Data Fig. 5). While this benchmark has the advantage of not being generated under any specific model, it offers less flexibility with respect to the dataset characteristics that could be varied and analyzed.

Analyzing the effect of different factors on DEBIAS-M's performance, we found similar performance with variable sequencing depths (1,000–100,000 reads per sample, Fig. 3a and Extended data Figs 4a,e,f, 5a). We have noted better performance with more batches (>2 in simulations and >3 on real data) and with more samples per batch (>40, Figs. 3b–c and Extended Data Figs. 4b–c, 5b–c). DEBIAS-M also demonstrated robustness to the strength of association with phenotype (Methods) and to the number of features (Figs. 3d–e and Extended Data Fig. 4d). Similarly, DEBIAS-M demonstrated consistent improvement compared to uncorrected abundances under increasing measurement noise (applied independently of bias; Methods; Fig. 3f), except when such noise severely distorted associations with phenotype (Extended Data Fig. 4g). Altogether, and despite the small variations in results across certain parameters, bias correction with DEBIAS-M produced microbial abundances that were more similar to the ground truth than those not corrected by DEBIAS-M across all ranges of simulated parameters (One-sided Wilcoxon signed-rank $p < 10^{-5}$ for all comparisons except one under the highest measurement noise; Fig. 3). Upon comparison to alternative methods, we have noted significant improvement using DEBIAS-M in both simulated and non-simulated settings in almost all evaluations, as well as improved prediction accuracy of simulated phenotypes (Extended Data Figs. 5, 6). We note, however, that as the simulated data in one of the benchmarks is generated under the DEBIAS-M model, differences in performance between DEBIAS-M and alternative methods in this case are likely an indicator of different assumptions rather than indication of performance. Overall, these results demonstrate that DEBIAS-M's performance is robust, suggesting that it could be used on datasets with a wide range of characteristics.

DEBIAS-M learns interpretable bias-correction factors

The interpretability of batch-correction came into the spotlight recently²⁷. It is important to understand the changes made to the data to better interpret the overall analysis. We therefore next evaluated whether the bias correction performed by DEBIAS-M is interpretable, and whether the correction factors learned could be assigned biological meaning. First, we evaluated whether the correction factors inferred for each batch are stable, which would be expected if they are mostly driven by experimental conditions that are consistent for every batch. Analyzing the HIV dataset, we applied DEBIAS-M to samples and labels from all studies except a randomly selected half of one held-out study. We then repeated the same procedure for the complementary half of the held-out study and compared the bias-correction factors learned by DEBIAS-M for each half, finding high consistency across halves of different studies (Pearson $R=0.59$, $p < 0.001$, Fig. 4a).

Next, we sought to find which experimental properties are associated with the inferred bias-correction factors. We therefore analyzed them with respect to technical parameters that we curated for every study in the HIV dataset (Supplementary Table 1). We found that DNA extraction kit was the most important factor driving the biases learned by DEBIAS-M, in line with previous results from a different dataset¹⁵, accounting for 43% of variance (PERMANOVA $p=0.002$; Fig. 4b–c). We also found strong associations between the inferred bias-correction factors and both 16S region and sample type (fecal, rectal swab, etc.), explaining an additional 27 and 14% of the variance in bias-correction factors, respectively ($p=0.021$ and $p=0.039$; Fig. 4b–c). While the type of extraction kit used was most strongly associated with the learned bias-correction factors, we found that the detection (i.e., presence or absence) of bacteria from different studies was instead associated with the choice of 16S region (Extended Data Fig. 7a–c), as previously described^{45,46}.

Finally, we investigated whether taxon-specific factors are associated with the inferred processing bias. Comparing Gram-positive and negative bacteria, we found a significant difference between their inferred biases in samples processed with QIAamp kits (Mann-Whitney U $p=0.002$, 0.076, 0.489 for QIAamp, MagNA, and MoBio, respectively; Fig. 4d), echoing results from highly standardized settings¹¹. We further noted higher variance in bias-correction factors for studies that used robotic rather than manual experimental processes (Mann-Whitney U $p=0.03$; Extended Data Fig. 7d). Other taxon-specific attributes, such as 16S copy number, had a much weaker association with bias-correction factors (Pearson $R=-0.10$, $p=0.024$; Extended Data Fig. 7e). Altogether, DEBIAS-M's ability to detect microbe- and experimental-specific factors, across highly variable studies with potentially many confounders, demonstrates the sensitivity of our approach and indicates that the inferred biases potentially reflect genuine biological factors driving differences between processing protocols.

Regression and multi-task DEBIAS-M improves metabolite predictions

The generative model of DEBIAS-M is flexible and can be extended for prediction of categorical or continuous variables (Fig. 5a; Methods). To evaluate application of DEBIAS-M for regression, we next designed a benchmark in which vaginal microbiome data is used to predict the levels of multiple metabolites measured from paired samples collected simultaneously^{47,48}. This is a challenging task, as many metabolites are likely to be affected by factors such as the host or environmental exposures^{47,49}. We used similar benchmarks as above to compare regression DEBIAS-M to alternative batch-correction methods in a cross-batch generalization setting (Methods). Predictions made using DEBIAS-M were significantly more accurate, with a median [IQR] cross-batch Spearman ρ of 0.23 [0.10–0.39] across 684 metabolites, compared to 0.06 [–0.05–0.18] across all alternative methods and 0 [–0.10–0.11] for uncorrected data (one-sided Wilcoxon signed-rank $p<10^{-10}$ for all comparisons; Fig. 5b).

Because the bias-correction factors inferred by DEBIAS-M are associated with experimental design, a single set of factors should generalize for prediction of multiple phenotype labels. To test this, we next evaluated DEBIAS-M in a multi-task setting, in which a single set of bias-correction factors is learned per study alongside multiple models that

predict different phenotypes – in this case, the levels of different metabolites (Fig. 5a; Methods). We compared multi-task DEBIAS-M (Methods) both to the single-task version and to MelonnPan⁵⁰, a method for predicting metabolite levels from microbiome data. We retrained MelonnPan with default parameters and without any batch correction and evaluated all three methods on 297 metabolites selected by MelonnPan (Methods). Multi-task DEBIAS-M outperformed both single-task DEBIAS-M and MelonnPan, with a median [IQR] cross-batch Spearman ρ of 0.30 [0.18–0.46], compared to 0.26 [0.11–0.46] and 0.24 [0.01–0.39], respectively (one-sided Wilcoxon signed-rank $p=9\times 10^{-4}$ and $p=10^{-7}$; Fig. 5c). Interestingly, DEBIAS-M and multi-task DEBIAS-M obtained similarly accurate predictions of metabolites that MelonnPan did not flag as “well-predicted”, with a median [IQR] cross-batch Spearman ρ of 0.21 [0.09–0.34] and 0.24 [0.12–0.36], respectively (Extended Data Fig. 8a). Results were also similar in a classification setting (Extended Data Fig. 8b). These results offer further evidence that the bias-correction factors learned by DEBIAS-M are not task-specific, and generally illustrate the potential of transfer learning for microbiome models.

DEBIAS-M preprocessing improves generalization of cancer prediction

Finally, we sought to evaluate DEBIAS-M’s utility in analyses using more expressive machine-learning algorithms. To this end, we reran our cross-cohort analysis associating the cervicovaginal microbiome with cervical neoplasia (Fig. 2c–d), except that this time we employed DEBIAS-M as a preprocessing step, whose output was then used to train a random forest model with hyperparameter tuning (Methods; Extended Data Fig. 1e). For prediction of cervical carcinoma, the random forest model did not consistently improve on results from the linear model (median [IQR] auROCs of 0.56 [0.52–0.76] and 0.58 [0.57–0.60], respectively; Fisher’s multiple comparisons of one-sided Delong tests $p=0.14$; Fig. 6a–b). Preprocessing with DEBIAS-M improved the predictive performance of both, but again with no significant advantage for using random forest models (auROC of 0.76 [0.53–0.80] and 0.74 [0.70–0.86] for DEBIAS-M preprocessing into random forest and linear models, $p=0.34$; Fig. 6a–b). The same was observed for prediction of cervical intraepithelial neoplasia. While a random forest model offered some improvement over a linear model (median [IQR] auROC of 0.52 [0.48–0.52] and 0.50 [0.42–0.51]; Wilcoxon $p=0.16$), using DEBIAS-M improved both (0.65 [0.56–0.66] and 0.65 [0.55–0.65] with and without random forest, respectively; $p=2.4\times 10^{-5}$ for comparing random forest models with and without DEBIAS-M; Fig. 6c; Supplementary Table 2). We note that DEBIAS-M increases the measured α diversity of cervicovaginal microbiome samples (Wilcoxon $p<10^{-20}$, Fig. 6d), although it maintained the same presence and absence of all taxa (Fig. 6e). These results demonstrate that bias removal may have a stronger effect on model generalization in a cross-study prediction setting than the selection of a particular machine learning algorithm.

We also examined a challenging scenario, and replicated a cross-study analysis of predicting immunotherapy response (12-month progression-free survival) from the gut microbiome of patients with melanoma, in which a limited generalizability of microbiome-based prediction across cohorts was previously concluded⁵¹. Analyzing data from six cohorts^{51–53} (N of 18–55 subjects per study, total of 194; Methods; Supplementary Table 6) we observed a small but consistent improvement across all cohorts when adding DEBIAS-M as a

preprocessing step to the pipeline used in the original study, and a strong improvement in the cohort of Peters et al.⁵³ (Extended Data Fig. 9a). As with cervical neoplasia, using random forest models did not improve on linear models (Extended Data Fig. 9a). Overall, these results demonstrate the utility of DEBIAS-M in realistic applications of cross-study machine-learning models.

Discussion

Sequencing-based measurements of microbial communities have demonstrated substantial variability between experimental approaches, laboratories, and even batches of the same study, making it challenging to replicate results across studies. To address these issues, we developed DEBIAS-M, a method for microbiome processing-bias correction. DEBIAS-M offers three notable benefits: First, it is based on a specific framework that stipulates taxon- and protocol-specific multiplicative biases¹⁵, and, as a result, infers bias-correction factors that are interpretable, robust, and consistent. Second, its design facilitates the development of generalizable machine-learning models, as it can operate without using the outcome labels and data of a test dataset. Third, in extensive benchmarks across more than 30 different studies, we compared DEBIAS-M to alternative methods and demonstrated that it consistently improved the generalization of microbiome-based prediction models to held-out studies or batches in diverse settings. This includes colorectal cancer, cervical neoplasia, HIV, and metabolome prediction, and using both metagenomics and 16S rRNA sequencing data. DEBIAS-M is available as an open-source package at <https://korem-lab.github.io/DEBIAS-M/>.

The multiplicative processing-bias framework introduced by McLaren et al.¹⁵ implies that even the same set of samples processed differently (e.g., using different extraction kits) might yield different microbial abundances and associations with phenotypes (Fig. 1a). This is even more likely when samples originate in different studies, with additional technical (e.g., storage time) and biological (diet, genetics) differences, underlining the challenge of bias and batch effects in microbiome studies. In our results, these challenging differences manifested in models that perform worse than chance when evaluated across studies, seen as $\text{auROC}s < 0.5$.

DEBIAS-M makes two important operational assumptions. The first is that samples from different batches are generally similar. This assumption is inherent to many batch-correction methods, and of course would not hold in some scenarios, such as an analysis of microbiome data from different body sites. The second assumption is that there exists some association between the ground-truth (unbiased) microbiome data and the provided phenotype, which is weakened by processing bias, and therefore can be improved via bias correction. We thus expect DEBIAS-M to work better with informative phenotypes. The combination of the two assumptions ensures that DEBIAS-M does not overfit to either the cross-batch similarity or the available phenotype labels, and, conversely, does not remove informative variability. We note, however, that the second assumption limits DEBIAS-M's utility to certain large-scale efforts aimed at estimating measurement variability¹¹, which typically include a small number of samples and lack an informative phenotype.

While previously developed batch-correction methods reduce batch effects, they are not grounded in a framework that relates them to specific processes that generate variability between batches, and might produce transformations that are hard to interpret^{25,27}. In many cases their output should likely be interpreted as a global transformation of the data rather than an attempt to quantify the abundances of specific microbes. In contrast, we demonstrate that bias-correction factors inferred by DEBIAS-M are linked to specific experimental properties and are therefore interpretable. We posit that this is due to the restrictions imposed on the correction performed by DEBIAS-M, limiting it to per-taxa multiplicative bias. While there likely are processes driving differences between batches that are not captured by multiplicative per-taxon biases, our results demonstrate that this restrictive approach is beneficial. Nevertheless, there might be specific scenarios in which a combination of approaches may be useful.

While many batch-correction methods were developed primarily for association testing, in practice, many are used for predictive modeling. This highlights a major difference with DEBIAS-M, which is designed to support the development and validation of models on held-out data. Several alternative methods require phenotypic variables for all data, even for held-out test sets, leading to overfitting and “leakage” of test information into the training data, thereby invalidating tests for generalization⁴³. Other methods are able to perform batch correction without the use of phenotypes, but require the consideration of features (microbial abundance data) from the entire dataset. While, contrary to others⁴³, we do not believe this necessarily constitutes information leakage, it does limit the translation of these models. Conversely, DEBIAS-M can handle missing data, and offers substantial flexibility with respect to available microbial and outcomes data, including an “online” version which does not require any test data until after the model is trained.

We offer a few insights and considerations for the use of DEBIAS-M. First, we do not attempt to identify experimental procedures that are “better” than others, and in fact posit that variability in methods makes it easier to infer biases and ensure generalizability of results. However, our analysis showing lower variance in bias-correction factors for manual processing suggests a benefit for this approach, which should be balanced with practical considerations. Second, we note that our simulation results indicate that DEBIAS-M operates better with more samples per batch and with informative phenotypes. These could serve as consideration for design of future studies, in addition to obtaining orthogonal measurements (e.g., qPCR, dilution series), which may be used to estimate bias directly^{15,54}. Third, as processing bias may distort relative abundances in a sample-specific manner even within a study¹⁵, there is an interest in addressing it. DEBIAS-M shows some promise in this direction (Supplementary Note 1; Extended Data Fig. 9b), and so do some methods for differential abundance testing⁵⁵. Fourth, while DEBIAS-M, by design, will not remove a taxon detected in a study or introduce an undetected taxon, this could be modified by the advanced user - which, in our analysis, leads to more uniform bias correction without a substantial effect on performance (Supplementary Note 2; Extended Data Fig. 10a–c). Last, as processing biases can differ even between strains belonging to the same species^{56,57}, we recommend applying DEBIAS-M at the lowest taxonomic level available, with aggregation performed after DEBIAS-M processing (Supplementary Note 3; Extended Data Fig. 10d–e).

As consideration of multiplicative bias shows promise for microbiome data analysis, future work could investigate ways with which to assign a bias factor to each processing step. In addition, positive controls and dilution series can also be incorporated by future studies as means to evaluate specific biases⁵⁴. It is also possible that DEBIAS-M could be combined with probabilistic microbiome processing models, such as SCRUB⁵⁸, with the intention of using as much information as possible during each. Lastly, there are other measurements outside of the microbiome field that are susceptible to processing biases, such as transcriptomics or metabolomics, and there may be an opportunity to adapt DEBIAS-M to such scenarios.

Methods

The DEBIAS-M model: processing bias

Consider a matrix $X \in \mathbb{Z}^{+n \times m}$ representing the number of reads originating in one of m taxa for each of n samples; a vector $A \in \mathbb{Z}^{+n}$, where each $A_i \in \{1, \dots, B\}$ denotes the batch sample i originates from, with B total batches; and a phenotype vector $Y \in \{0,1\}^n$ that describes some information of interest that is expected to be associated with X .

The reads observed in each sample X_i are the result of some experimental process (e.g., DNA extraction, 16S rRNA gene amplification, and sequencing). This process attempts to measure the underlying “true” composition of m taxa in that sample, which we denote $\Gamma_i \in \mathbb{R}^m$. Γ_i represents relative abundances, such that all $\Gamma_{i,j} \geq 0$ and $\sum_j \Gamma_{i,j} = 1$. As proposed by McLaren et al.¹⁵, for each batch $b \in \{1, \dots, B\}$ and taxon j we assume a specific multiplicative bias term that can increase or decrease the likelihood of $\Gamma_{i,j}$ to be observed in the downstream X_i . These biases can be due to interbacterial differences in DNA yield, gene copies, PCR primers, extraction protocols, and so on. Since the biases inflicted by each experimental processing stage are all assumed to be multiplicative, they can be aggregated into a single bias factor per microbe within every batch¹⁵. To capture this phenomenon, we propose one weighting parameter $W_{A,j} > 0$ for each taxon-batch pair, where each $W_{A,j}$ represents the multiplicative bias that protocol b had on taxon j . Thus, we draw each observed sample X_i from a multinomial mixture $X_i \sim \text{Multinomial}\left(C_i, \left(\frac{W_{A_i,1}\Gamma_{i,1}}{\sum_{o=1}^m W_{A_i,o}\Gamma_{i,o}}, \frac{W_{A_i,2}\Gamma_{i,2}}{\sum_{o=1}^m W_{A_i,o}\Gamma_{i,o}}, \dots, \frac{W_{A_i,m}\Gamma_{i,m}}{\sum_{o=1}^m W_{A_i,o}\Gamma_{i,o}}\right)\right)$, where $C_i = \sum_{j=1}^m X_{i,j}$ represents the total number of reads observed in sample X_i . So, for any sample i , the expected observed number of reads observed from a taxon j is:

$$\mathbf{E}[X_{i,j} | \Gamma, \mathbf{W}] = C_i \frac{W_{A_i,j}\Gamma_{i,j}}{\sum_{o=1}^m W_{A_i,o}\Gamma_{i,o}}$$

Solving for $\Gamma_{i,j}$, and, for ease of notation, denoting $Z_{i,j} = \frac{\mathbf{E}[X_{i,j}]}{C_i}$ as the expected relative abundance of a taxon in a sample, we find:

$$\Gamma_{i,j} = \frac{1}{W_{A_i,j}} Z_{i,j} \left(\sum_{o=1}^m W_{A_i,o} \Gamma_{i,o} \right) \quad \#(1)$$

By construction, each Γ_i represents a relative abundance space, and therefore $\sum_{j=1}^m \Gamma_{i,j} = 1, \forall i$. Substituting (1) into this relative abundance constraint:

$$\sum_{j=1}^m \left[\frac{1}{W_{A_i,j}} Z_{i,j} \left(\sum_{o=1}^m W_{A_i,o} \Gamma_{i,o} \right) \right] = 1$$

Factoring out the summations, and solving for the inner summation:

$$\sum_{o=1}^m W_{A_i,o} \Gamma_{i,o} = \frac{1}{\sum_{j=1}^m \frac{1}{W_{A_i,j}} Z_{i,j}}$$

Substituting the right-hand side back into (1) (note a change in the denominator's indicator variable):

$$\Gamma_{i,j} = \frac{\frac{1}{W_{A_i,j}} Z_{i,j}}{\sum_{o=1}^m \frac{1}{W_{A_i,o}} Z_{i,o}}$$

For simpler notation, denote $W'_{A_i,j} = \frac{1}{W_{A_i,j}}$, and note that since W is the processing bias, W' could be interpreted as a “bias-correction factor”. Using the fact that the constant denominators cancel out across the $Z_{i,j}$ terms, we obtain the expected value of Γ , which corresponds to the $\hat{\Gamma}_{i,j}$ estimate of DEBIAS-M:

$$\mathbb{E}[\Gamma_{i,j} | X, W'] = \hat{\Gamma}_{i,j} = \frac{W'_{A_i,j} X_{i,j}}{\sum_o W'_{A_i,o} X_{i,o}} \quad \#(2)$$

$\hat{\Gamma}$ are the de-biased microbial abundances, which, after estimation of W' , will be part of the output of DEBIAS-M.

The DEBIAS-M model: phenotype associations

Next, we assume that there exists an association between the expected $\hat{\Gamma}$ and the phenotype Y . While DEBIAS-M can model this association in various ways, we first describe logistic regression, with a single set of linear weights $L \in \mathbb{R}^{m+1}$, such that $Y | X, W, L \sim \text{Bernoulli}(Y, \sigma(L\hat{\Gamma}))$, where σ is the sigmoid function. The first m columns correspond to the number of microbes, while column $m+1$ is a constant term for the

intercept, such that $\hat{T}_{i,m+1} = 1, \forall i$. We assume that in the absence of processing bias, the association between $\hat{T}$ and Y is protocol- and batch-independent, and therefore the weights L should be identical across all batches. Given this, the probability of observing a set of labels Y given $\hat{T}$ and L is:

$$\mathbf{P}(Y|X, \mathbf{W}, L) = \prod_{i=1}^n \left([\sigma(L\hat{T}_i)]^{Y_i} \cdot [1 - \sigma(L\hat{T}_i)]^{1-Y_i} \right)$$

The DEBIAS-M model: cross-batch similarity

We next make an additional assumption, that batches of the same sample types in similar contexts should be similar, as is standard for established batch-correction methods^{16,18–20}. This would generally apply to microbiome data collected from the same type of environment (e.g. human gut from studies sharing the same patient exclusion criteria). We express this assumption by comparing the L^2 distance between the pairwise average of the $\hat{T}$ inferred for each batch, and note that the constant $m+1$ -th column of $\hat{T}$ is ignored here. Thus, we introduce $\mu \in \mathbb{R}^{B \times m} = \mu(\hat{T}, A)$, the average of each batch:

$$\mu_{b,j} = \frac{\sum_{i|A_i=b} \hat{T}_{i,j}}{|i|A_i=b|}$$

We assume there exists an underlying probability in which batch means μ_{b_1} and μ_{b_2} are equivalent. Then, assuming that each μ follows a multivariate Gaussian distribution which we set with a covariance matrix of $\Sigma = \epsilon I$, where I is the identity matrix and ϵ is a scaling hyperparameter, such that all $\mu_b \sim \mathcal{N}(\mu', \epsilon I)$, we offer a simplification to directly optimize the pairwise L^2 penalty in log space. This would amount to the following expression if we consider each pairwise batch averages as a variable/mean-parameter combination:

$$\mathbf{P}(X, A|\mathbf{W}, \epsilon) = \prod_{j=1}^m \prod_{b_1=1}^B \prod_{b_2=1}^B \left(\frac{1}{\epsilon\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{\mu_{b_1,j} - \mu_{b_2,j}}{\epsilon} \right)^2 \right] \right)$$

We use this approach to produce an L^2 penalty, but note that a similar approach could be used to produce an L^1 penalty. We further note that the DEBIAS-M software package enables to weight this loss by study size, although in our benchmarks this did not result in improved accuracy.

Adding this cross-batch similarity term to the prediction component yields:

$$\begin{aligned} \mathbf{P}(Y, X, A|\mathbf{W}, L, \epsilon) &= \mathbf{P}(Y|X, \mathbf{W}, L) \cdot \mathbf{P}(X, A|\mathbf{W}, \epsilon) = \prod_{i=1}^n \left([\sigma(L\hat{T}_i)]^{Y_i} \cdot [1 - \sigma(L\hat{T}_i)]^{1-Y_i} \right) \\ &\cdot \prod_{j=1}^m \prod_{b_1=1}^B \prod_{b_2=1}^B \left(\frac{1}{\epsilon\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{\mu_{b_1,j} - \mu_{b_2,j}}{\epsilon} \right)^2 \right] \right) \end{aligned}$$

We note that this cross-batch similarity assumption assumes only a similarity in means, and is thus weaker than the similarity assumption made by some alternative methods, which enforce similarity on both mean and variance^{20,22,23} or on the entire shape of the distribution using quantiles^{16,18,19,21}. We further note that cross-batch similarity is not applied alone, but together with prediction loss, penalizing DEBIAS-M for removal of meaningful phenotypic variation.

The DEBIAS-M model: optimization

The DEBIAS-M model is implemented by maximizing the log of the combination of the two components described above, with respect to the parameters L and W' , which represents an optimization problem utilizing the observed X, Y , and A . Additionally, note that the ϵ hyperparameter, along with some constants, is absorbed into the hyperparameter λ . More explicitly, we denote:

$$\log P(X, Y, A | W', L, \lambda) = \sum_{i=1}^n \left(Y_i \log(\sigma(L\hat{T}_i)) + (1 - Y_i) \log(1 - \sigma(L\hat{T}_i)) \right) - \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2 \quad \#(3)$$

Where:

$$\hat{T}_{i,j} = \frac{W'_{A_i,j} X_{i,j}}{\sum_{o=1}^m W'_{A_i,o} X_{i,o}} \text{ if } j \leq m, 1 \text{ if } j = m+1$$

and:

$$\mu_{b,j} = \frac{\sum_{i|A_i=b} \hat{T}_{i,j}}{|\{i|A_i=b\}|}$$

We note that the cross-batch similarity term $-\lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2$ is not convex with respect to W' because the means are compositionally scaled (as seen in equation (2)), and therefore, equation (3) is not convex.

Using this expression for $\log P(X, Y, A | W', L, \lambda)$, we iteratively backpropagate all the way through the $\hat{T}_{i,j}$ estimate to identify which W', L parameters maximize the log-likelihood.

In practical applications of batch correction, there are often samples or labels that are designated as a test set and are therefore hidden from the model (Extended Data Fig. 1). We first consider a common scenario for batch-correction (Extended Data Fig. 1a), in which the features (microbial counts) for all samples (including the test set) are available during training, but there exists a set of points $\mathcal{T}$ (the test and validation sets), for which the Y_i labels are unavailable for all $i \in \mathcal{T}$. For this scenario, the optimization to obtain the fitted parameters is:

$$L^*, W^* = \operatorname{argmax}_{L, W} \sum_{i \notin \mathcal{T}}^n (Y_i \log(\sigma(L\hat{T}_i)) + (1 - Y_i) \log(1 - \sigma(L\hat{T}_i))) - \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2$$

where the μ and $\hat{T}$ terms are as described above. Through this optimization, DEBIAS-M can infer the bias correction factor W' parameters that yield the corrected sample compositions, inferred in equation (2). While the multinomial derivation was described for input in count space, we note that the same optimization equations would work for input $X \in \mathbb{R}^{n \times m}$ representing relative abundances, as the $E[\Gamma_{i,j}]$ step in equation (2) will still produce estimates in relative abundance spaces, thus enabling the same optimization process.

Online DEBIAS-M

In some scenarios, neither the features (microbial counts) nor the labels are available for the test set. In this case, we separate the modeling into two phases. The first, denoted as the training phase, is implemented to infer the prediction weights L and the W' bias-correction factors for all batches available during training. Denote $\Omega = \{A_i | i \notin \mathcal{T}\}^c$ as the set of batches comprising only test samples, which are completely unavailable during training (note that we describe the set as the complement of batches that include training samples). Therefore, during the training stage, we focus only on the batches that include training samples:

$$\mu_{b,j} = \frac{\sum_{i|A_i=b, i \notin \mathcal{T}} \hat{T}_{i,j}}{|\{i | A_i=b, i \notin \mathcal{T}\}|}, \forall b \notin \Omega$$

We emphasize that these batch averages correspond to the debiased $\hat{T}_{i,j}$ estimates of only the training samples. These components are used to obtain estimates for the predictive model's L^* , and for the bias-correction factors W^* only from within the training sets, by maximizing the following equation:

$$L^*, W_{b \notin \Omega}^* = \operatorname{argmax}_{L, W_{b \notin \Omega}} \sum_{i \notin \mathcal{T}}^n (Y_i \log(\sigma(L\hat{T}_i)) + (1 - Y_i) \log(1 - \sigma(L\hat{T}_i))) - \lambda \sum_{j=1}^m \sum_{b_1 \notin \Omega} \sum_{b_2 \notin \Omega} (\mu_{b_1,j} - \mu_{b_2,j})^2$$

We emphasize that samples from the testing set Ω are used in neither the prediction loss nor the cross-batch similarity. Also, note that from this point on, the prediction model L^* and training batch bias-correction factors $W_{b \notin \Omega}^*$ are frozen, and not updated or adjusted in any way once the training step has concluded. Next, once data from a test set is received, the online adaptation step infers just the bias correction factors from within the test batch (or batches) Ω , which corresponds to only optimizing the W^* parameters from those batches $W_b, \forall b \in \Omega$. This requires inferring the batch averages using the test set samples (from Ω), which corresponds to the following μ terms:

$$\mu_{b,j} = \frac{\sum_{i|A_i=b, i \in \mathcal{T}} \hat{T}_{i,j}}{|\{i | A_i=b, i \in \mathcal{T}\}|}, \forall b \in \Omega$$

Then, the optimization to obtain parameter estimates for the new test batches' bias-correction factors W_b , $\forall b \in \Omega$ is minimizing the observing differences with the μ terms of the frozen training-set batches, which do not change during this step because the $W_{b \notin \Omega}^*$ are frozen. We are therefore only adjusting the bias factors for the testing batches $W_{b \in \Omega}^*$:

$$W_{b \in \Omega}^* = \underset{W_{b \in \Omega}}{\operatorname{argmax}} \left(- \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2 \right)$$

Within all batches, the debias-ed sample compositions $\hat{T}_{i,j}$ are always determined by the product of the corresponding samples' bias-correction factors W^* and the observed sample abundances, re-scaled into relative abundance space:

$$\hat{T}_{i,j} = \frac{W_{A_{i,j}}^* X_{i,j}}{\sum_{o=1}^m W_{A_{i,o}}^* X_{i,o}}$$

The predictions are made using the already-frozen L^* s applied to the test-set $\hat{T}$ s without any retraining.

Log-additive DEBIAS-M

As it is common and effective to analyze microbiome datasets in log-space, we also provide a version of DEBIAS-M tailored to this feature space. As bias terms are multiplicative in count and relative abundance space, they are additive in log space. We therefore refer to this version of DEBIAS-M as 'log-additive DEBIAS-M' (used in Extended Data Figs. 2d–g).

In clr-transformed space, the unbiased $\hat{T}_{i,j}$ term can be represented with the following expression, which includes normalization that mimics the same renormalization process in relative abundance space:

$$\hat{T}_{i,j} = \frac{W_{A_{i,j}}' + X_{i,j}}{\sum_{o=1}^m |W_{A_{i,o}}' + X_{i,o}|}$$

We note that different logarithmic transformations would likely require different normalization terms. As before, we add a constant $m+1$ column $\hat{T}_{i,m+1} = 1$, $\forall i$ to accommodate the constant parameter in the predictive models.

Regression DEBIAS-M

To use DEBIAS-M for regression, in which all $Y_i \in \mathbb{R}$ rather than binary, we consider the same parameters for optimization, but replace the sigmoid function and classification loss with regression loss using the sum of squared error, as follows:

$$L^*, W^* = \underset{L, W}{\operatorname{argmin}} \sum_{i \notin \mathcal{T}}^n (Y_i - L\hat{T}_i)^2 + \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2$$

Multitask DEBIAS-M

This variation of DEBIAS-M supports multiple phenotypes per sample, which we show to assist in inference. We represent this by considering an integer T which represents the number of distinct training tasks available, such that $Y \in \{0,1\}^{n \times T}$, where $Y_{i,t}$ represents the label of sample i for phenotype t . Corresponding to these multiple tasks, we now consider T sets of linear weights $L_1, L_2, \dots, L_T$, such that $L \in \mathbb{R}^{T \times (m+1)}$. Under this scenario, the optimization for DEBIAS-M is:

$$L^*, W^* = \operatorname{argmax}_{L, W} \sum_{t=1}^T \sum_{i \notin \mathcal{T}}^n (Y_{i,t} \log(\sigma(L_t \hat{\Gamma}_i)) + (1 - Y_{i,t}) \log(1 - \sigma(L_t \hat{\Gamma}_i))) - \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2$$

And note that with $T = 1$ this corresponds to a single-task DEBIAS-M.

This multi-task approach also generalizes to regression, which we demonstrate below for a set of regression tasks such that all $Y_{i,t} \in \mathbb{R}$:

$$L^*, W^* = \operatorname{argmin}_{L, W} \sum_{t=1}^T \sum_{i \notin \mathcal{T}}^n (Y_{i,t} - L_t \hat{\Gamma}_i)^2 + \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2$$

A general DEBIAS-M optimization scheme

More broadly, for any loss function $\mathcal{L}(y, f(\hat{\Gamma}|L))$, where y can represent either classification or regression labels, and $f(\hat{\Gamma}|L)$ is a prediction function targeting y , given some relative abundance estimate $\hat{\Gamma}$ and parameters L , we can represent any DEBIAS-M model as:

$$L^*, W^* = \operatorname{argmin}_{L, W} \sum_{t=1}^T \sum_{i \notin \mathcal{T}}^n \mathcal{L}(Y_{i,t}, f(Y_{i,t}, \hat{\Gamma}_i | L_t)) + \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2$$

To evaluate the effect of different loss functions used for classification, we compared their performance on the HIV and colorectal cancer benchmarks. While we have not found consistent differences (Extended Data Fig. 10f–g), we do find data-set specific differences, making loss function a hyperparameter the advanced user may wish to optimize.

L^2 regularization of DEBIAS-M

To reduce the risk of overfitting, DEBIAS-M also supports regularization terms. To this end, we introduce two additional hyperparameters, λ_1 and λ_2 ($\lambda_1 \geq 0$ and $\lambda_2 \geq 0$), which correspond to the L^2 regularization weights for the bias and predictive coefficients, respectively. Incorporating these regularization terms into the general DEBIAS-M optimization scheme, we obtain the following:

$$L^*, W^* = \underset{L, W}{\operatorname{argmin}} \sum_{t=1}^T \sum_{i \notin \mathcal{I}}^n \mathcal{L}(Y_{i,t}, f(Y_{i,t}, \hat{\Gamma}_i | L_t)) + \lambda \sum_{j=1}^m \sum_{b_1=1}^B \sum_{b_2=1}^B (\mu_{b_1,j} - \mu_{b_2,j})^2 \\ + \lambda_1 \sum_{b=1}^B \sum_{j=1}^m (\log_2(W'_{b,j}))^2 + \lambda_2 \sum_{t=1}^T \sum_{j=1}^m (L_{t,j})^2$$

While DEBIAS-M offers these options, we have set λ_1 and λ_2 to zero by default, except in Extended Data Fig. 9a where we tuned these hyperparameters using k-fold cross validation within the training studies.

Implementation of DEBIAS-M

DEBIAS-M is implemented via batch gradient descent through pytorch-lightning, using the adam optimizer with a learning rate of 0.005 and otherwise default parameters, run with at least 25 epochs. Before the DEBIAS-M optimization, the linear weights L are initialized using a scikit-learn LogisticRegression model trained using the uncorrected data. The $\log_2(W)$ are stored as free parameters, thus ensuring that all the W are positive. These log-parameters are initialized to 0, corresponding to multiplicative bias-correction factor W' of 1. While the hyperparameter λ can be tuned via cross-validation, we found that DEBIAS-M performs well when it is weighted as a function of the number of features and possible pairwise batch comparisons. We therefore use $\lambda = \lambda(m, B) = \frac{\lambda'}{m \cdot B(B-1)/2}$ and set $\lambda' = 10^4$ by default. We empirically found that 10^4 is an effective scaling value; this expression was used as our λ' weight for all analyses. For log-additive DEBIAS-M, we modified the λ' hyperparameter to 10^3 , to account for the larger range of values in the clr-transformed data.

All implementations and per-model training of DEBIAS-M used in this work required less than one minute of runtime on a standard laptop. To evaluate how this runtime scales with dataset size, we have performed two benchmarks: in the first, we simulated datasets with an increasing number of batches, each containing 96 samples. In the second, we simulated datasets with 8 batches, but with an increasing number of samples per batch. While the runtime of DEBIAS-M did scale exponentially with the number of batches, it ran in <15 minutes (median 187 seconds) on a standard laptop even for absurdly large datasets (64 batches, i.e., 6,144 samples; Extended Data Fig. 10h). We observed a similar trend when increasing the number of samples with a fixed number of batches: in this case, DEBIAS-M runs in <43 seconds (median 33 seconds) for 512 samples per batch (a total of 4,096 samples; Extended Data Fig. 10i).

Microbiome data acquisition and processing

All datasets used in this work were publicly available at the time of analysis. We obtained the HIV data²⁸ from Synapse (<https://www.synapse.org/#!Synapse:syn18406854>), using the “taxonomic_assignments/insight.merged_otus.txt” file with data processed using Resphera Insight²⁸. We obtained the cervical neoplasia datasets from the repository available for each study (Supplementary Tables 4,5), and processed it with DADA2⁵⁹. We used any

indication of cervical intraepithelial neoplasia as phenotype in its analysis. Note that due to small sample size of the cervical carcinoma datasets (Supplementary Table 5), the cervical intraepithelial neoplasia datasets (Supplementary Table 4) were also used for training. The colorectal cancer data was obtained from the R curatedMetagenomicData package⁶⁰, which provides species-level relative abundance data processed by MetaPhlAn3. The paired vaginal microbiome and metabolite data were obtained from Supplementary Data 2 of ref.⁴⁸ and Supplementary Table 1 of ref.⁴⁷ Bacterial metadata was obtained from <https://gold-ws.jgi.doe.gov/>.

Benchmarking of batch-correction methods

For all benchmarks we ran: (1) ComBat using the “ComBat_seq” function applied to count data with an added pseudocount; (2) ConQuR using the “ConQuR” function applied to count data; (3) percentile normalization using the “percentile_norm” function from the PLSDAbatch package, applied to relative abundance data, with outputted NaNs replaced to 10^{-6} ; (4) voom-SNM using code made available by Poore et al. (file “Plasma-Voom-SNM-Normalize-Age-and-Sex.R”)²⁶ applied to count data; (5) MMUPHin using the package’s “adjust_batch” function applied to relative abundance data; and (6) PLSDA_batch using the “PLSDA_batch” function applied to relative abundance data. As ConQuR, percentile normalization, and Voom-SNM require a covariate variable and do not withhold that variable from the test set, we used age for the HIV dataset, gender for the colorectal cancer studies, age > median for the cervical neoplasia datasets, and self-identified race for the metabolites dataset. Note, however, that in Extended Data Fig. 6 these methods received the phenotype label for all batches. Most datasets did not have more than one covariate with values for most samples; for the HIV dataset we had a choice of age and gender, and there was no significant difference in performance between them (Extended Data Fig. 3c). The outputs of Voom-SNM, ConQuR, Combat, percentile normalization, MMUPHin, and PLSDA-batch, along with uncorrected (raw) relative abundances, were assessed using logistic regression with no penalty. For DEBIAS-M, the correction and prediction are implemented simultaneously through a similarly unregularized linear layer, without considering any metadata except for the outcome label of the training data. Of note, such separation between training and testing data is not available for other batch-correction methods. In benchmarks involving log-additive DEBIAS-M (Extended Data Fig. 2d–g), we used similar LogisticRegression models. PLSDA-batch can run on centered-log-ratio (clr)-transformed abundances²⁰, while for other methods that produce output in count or relative abundance space, we clr-transformed their output. We omitted Voom-SNM from this analysis because it cannot operate on clr-transformed values and its output is not in count or relative abundance space.

We implemented a cross-study validation pipeline, in which we trained a model on data from all but one study, and evaluated the predictive performance of the model on the held-out study, such that in the boxplots in Figs. 2, 3, 6a–c and Extended Data Figs. 2c–g, 3, 4a–f, 5, 6, 9a, and 10c–g each “dot” represents that held-out study (for all studies in the dataset). To compare the performance of multiple classifiers on the same prediction benchmark (e.g. HIV classification following DEBIAS-M and an alternative method) we used Fisher’s multiple

comparison to combine one-sided DeLong tests performed on each pairwise comparison of performance on the same held-out study.

For the prediction benchmarks in Fig. 6, we ran the same general predictive pipeline for cervical carcinoma and cervical intraepithelial neoplasia, but rather than using a logistic regression model, we used a random forest model. For this model, we tuned the max depth and max features hyperparameters using cross-validation on 3 folds (nested within training data). For the prediction benchmark in Extended Data Fig. 9a, we considered both the same random forest tuning pipeline, and an L^1 logistic regression model of the relative abundance data in log space with a pseudocount of 10^{-4} , as in the original analysis⁵¹. In both cases, we also tuned the hyperparameters of DEBIAS-M (with nested cross-validation): the learning rate, L^2 regularization strength, and λ' (using 10^3 , 10^4 , and 10^5).

Simulation of synthetic processing biases

To evaluate the robustness of the DEBIAS-M model across different scenarios, we used data-driven simulations to measure it against a known ground truth. To this end, we used the pretrained gut sample generator of SparseDOSSA2⁴⁴ to generate simulated data, using default parameters. With this trained model, we generated 25 synthetic datasets of 100 features and 384 samples, from which we created simulations of processing biases and batch effects. To simulate bias, we drew $\log_2$ bias factors for each study-feature combination from i.i.d Gaussian distributions of mean 0 and standard deviation of 2. The exponents of these factors were then multiplied by the taxonomic abundance vectors of each sample from the corresponding study, before being proportionally renormalized to simulate sequencing depths of 10^3 , 10^4 , or 10^5 . To simulate phenotype labels, we randomly generated linear weights for each feature, drawn i.i.d from a Gaussian distribution of mean 0 and standard deviation of 2. Multiplying these simulated linear weights by each sample yielded a score for each sample, which is by construction perfectly associated with the simulated microbiome. We then modulated the strength of this association by adding noise drawn i.i.d from a Gaussian distribution with a mean of zero and standard deviation of 0.1, 1, or 10. These were used to generate the final phenotype label – with “strong”, “moderate”, and “weak” associations, respectively – which were binarized using the median as a threshold. The final labels corresponded to average PERMANOVA R^2 with the simulated microbiome data of 0.086 (median $p \leq 10^{-5}$), 0.041 (median $p \leq 10^{-5}$), and 0.007 (median $p = 0.23$) for “strong”, “moderate”, and “weak”, respectively.

Overall, our simulations varied: 1) sequencing depth, as detailed above; 2) the number of batches, using 2, 3, or 4 batches; 3) the numbers of samples per batch, using 24, 48, 72, or 96 samples; 4) the strength of association with a simulated phenotype (weak, moderate, or strong), as detailed above; 5) the number of features, using 10 or 100 features; and 6) adding additional measurement noise, as detailed below. Whenever a factor was varied, all other factors were kept constant, using 4 studies, 96 samples per study, 100 features, strong phenotype associations, a measurement noise of zero, and read depth of 10^4 .

For sequencing depth, our main simulations (Fig. 3a) use the same depth across all samples, simulating a single rarefaction threshold. We also performed a variation of this

analysis in which a different depth was drawn for each batch (drawn from a log-normal distribution with a mean of 10^3 , 10^4 , or 10^5 reads; Extended Data Fig. 4e), which simulates rarefaction to a batch-specific threshold; and another variation in which each sample had a different depth, drawn from a log-normal distribution with a batch-specific mean (which was similarly drawn from a log-normal distribution; Extended Data Fig. 4f).

For measurement noise (Fig. 3f), we added noise in relative abundance space to the generated biased samples. Measurement noise was drawn from uniform distributions ranging from zero to the specified noise level, with different draws added into any nonzero feature within each sample. To evaluate the impact of each specified measurement noise level on the true signal between the samples and the labels, we performed bias correction of the noisy samples using the simulated ground-truth bias correction factors, followed by prediction scoring using the simulated ground-truth prediction parameters. These recreated phenotype scores using noisy samples were compared against the ground-truth phenotype scores before the binarization step (Extended Data Fig. 4g).

For every experiment, all biased samples were provided to DEBIAS-M, along with the simulated labels from the training set. We then compared the debiased data provided by DEBIAS-M for the held-out test set (for which no labels were provided) against the simulated ground-truth via Jensen-Shannon divergence (JSD). The same comparison was also made for the uncorrected samples. The median JSD for each experiment was then recorded, for a total of 25 points per box in Fig. 3. Within this simulation framework, we also evaluated the DEBIAS-M fitted bias correction factors against the simulated ground-truth biases for features that were nonzero in at least 10% of the dataset, again evaluating on a held-out test batch (Extended Data Fig. 4a–d).

We additionally ran simulations (Extended Data Fig. 5) using the colorectal cancer dataset (analyzed in Fig. 2b), in a benchmark evaluating cross-study generalization. In this analysis, we synthetically changed the benchmark to use either fewer reads per sample (10^3 , 10^4 , or 10^5), fewer studies (3, 6, or 9), or fewer samples per study (up to 20, 40, 60, 80, or 100 - we did not exclude studies not meeting this threshold). Results were evaluated via the area under the ROC curve of the predictions on the held-out test study, as in Fig. 2b. Note that this benchmark makes no assumptions on the data-generating process.

To compare DEBIAS-M to other methods on our simulated benchmarks (Extended Data Fig. 6a–f), we ran alternative methods that output transformed data in either counts or relative abundance space (ComBat, ConQuR, percnorm, and MMUPHin) on the same simulations as in Fig. 3. In this analysis, methods that require the outcome label (ConQuR, percnorm) were provided with the simulated labels for all samples, including the test batch (DEBIAS-M was only provided with the simulated labels for the training batches, as described above). We evaluated the performance of all methods by calculating the JSD between the corrected samples in the test set and the simulated ground truth of those samples. Additionally, to evaluate these simulations in a predictive setting, we evaluated the cross-study prediction generalization of each method using the same approach as in Extended Data Fig. 1a–b, along with the prediction evaluation of models trained and tested on the known ‘true’ sample relative abundances. In this analysis, ConQuR and percnorm were provided with a covariate

that we generated similarly to the phenotype label, which was used as in Extended Data Fig. 1b.

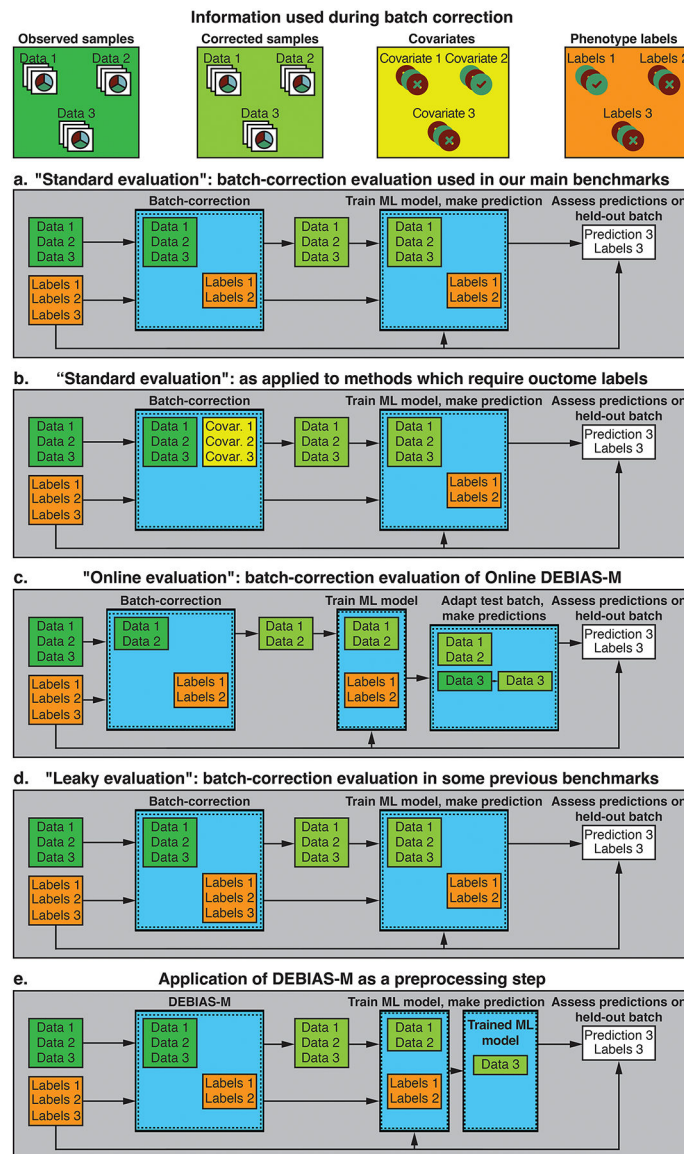
Inference of DEBIAS-M bias-correction factors

To investigate the bias-correction factors inferred by DEBIAS-M, we utilized the collection of HIV datasets, which included studies with a wide range of experimental designs. We began by implementing DEBIAS-M using our standard train-test split, but running it twice per validation batch, with each iteration observing a randomly selected half of the batch. The resulting inferred bias correction factors for OTUs that were present in both semi-batches were compared in aggregate across all studies (Fig. 4a). Next, we evaluated a DEBIAS-M model that included the samples and labels of all batches. While in the analysis in Fig. 4 bias-correction factors for non-detected taxa were kept at 1, in Extended Data Fig. 7 we instead imputed them to the largest observed bias-correction factor, implying an assumption that maximal bias against the taxa caused it to go undetected. We also evaluated the detection (presence/absence) of each taxa in the HIV studies directly, and evaluated these differences via agglomerative clustering, using the Manhattan distance metric and the complete linkage method. Gram status and 16S copy number were obtained from the GOLD database⁶¹.

Analysis of regression and multi-task DEBIAS-M

The benchmark for prediction of metabolite abundance used the metabolite data and metadata from Table S1 in Kindschuh et al.⁴⁷, and the microbiome data from Supplementary Data 2 in Elovitz et al.⁴⁸, which was processed in two batches. For each of the 684 metabolites with at least 50 measurements, we evaluated models predicting the level of a metabolite, as measured by mass-spectrometry⁴⁷, from microbiome data. We used the larger batch as training data and the smaller one as a test set. For all methods, the metabolite levels were adjusted into log-space, followed by standard scaling. We used MelonnPan⁵⁰ on the raw data within the training set with standard parameters and the most lenient threshold for selecting metabolites for prediction. As required by MelonnPan, we provided it with the metabolite levels normalized into relative abundance space rather than standard scaling. MelonnPan selected 297 metabolites, whose predictions were also evaluated using the single-task DEBIAS-M and the Multi-task DEBIAS-M which jointly considered all metabolites (Fig. 5c). We also evaluated the predictions of Multi-task DEBIAS-M for metabolites observed in a minimum of 50 samples which were not selected for prediction by MelonnPan (Extended Data Fig. 8a). For all regression predictions, performance was measured using Spearman correlation; in cases where a model predicted constant values for all test samples, we considered the correlation to be zero. We also ran Single- and Multi-task DEBIAS-M for metabolite classification, to determine if a particular sample had a measured metabolite level greater than the dataset's median, for the metabolites which had multiple observed values in both processing batches following this label processing and were selected by MelonnPan. Regression predictions by MelonnPan were evaluated on these classification labels via auROC, as were the DEBIAS-M predictions (Extended Data Fig. 8b).

Extended Data

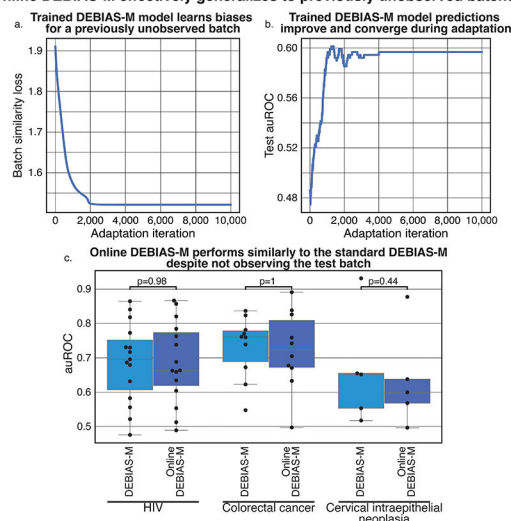


Extended Data Figure 1 |. Overview of information used in various microbiome batch-correction prediction benchmarks.

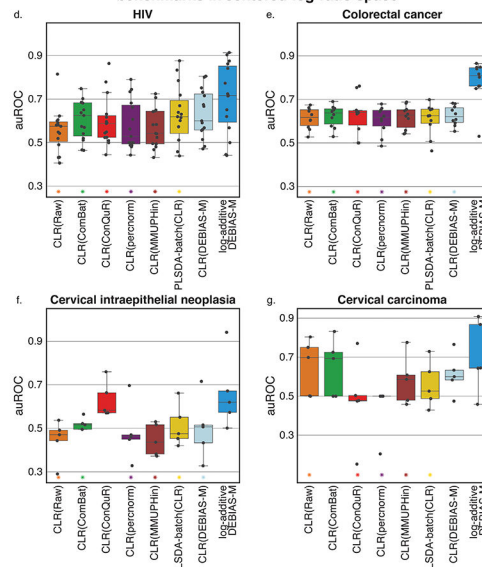
Top legend: Description of information that is typically incorporated in microbiome batch correction, which is 1) the samples themselves; 2) the labels to be predicted in downstream modeling; 3) other covariates; and 4) target labels. **a**, The primary batch-correction evaluation strategy used in this work for DEBIAS-M, in which the samples from all studies are used during batch correction, but only the labels from the training set are used during batch correction or model training. **b**, The primary batch-correction evaluation strategy used for methods that require an outcome label for all studies, in which a covariate is provided during batch-correction instead of the target label. **c**, The batch-correction strategy used in our 'online' benchmark in Extended Data Fig. 2a–c, in which no information from the test set is used during batch correction or model training. Once all bias-correction factors

and predictive model weights are learned and fixed for the training set, bias correction is performed separately for the test set by adjusting its bias-correction factors to optimize cross-batch similarity. **d**, An approach that has been used in some previous benchmarks, in which the labels of the test set are used during batch correction itself. This risks “information leakage”, and is used in this work only in Extended Data Fig. 3a–b, and, for ConQuR and percnorm, also Extended Data Fig. 6. **e**, A description of how DEBIAS-M is used as a preprocessing step (Figs. 6 and Extended Data Fig. 9a). DEBIAS-M is applied to samples from all studies, but receives the target label only for studies from the training set. A prediction model is then trained on the training set which was processed by DEBIAS-M, and evaluated on the test set.

Online DEBIAS-M effectively generalizes to previously unobserved batches

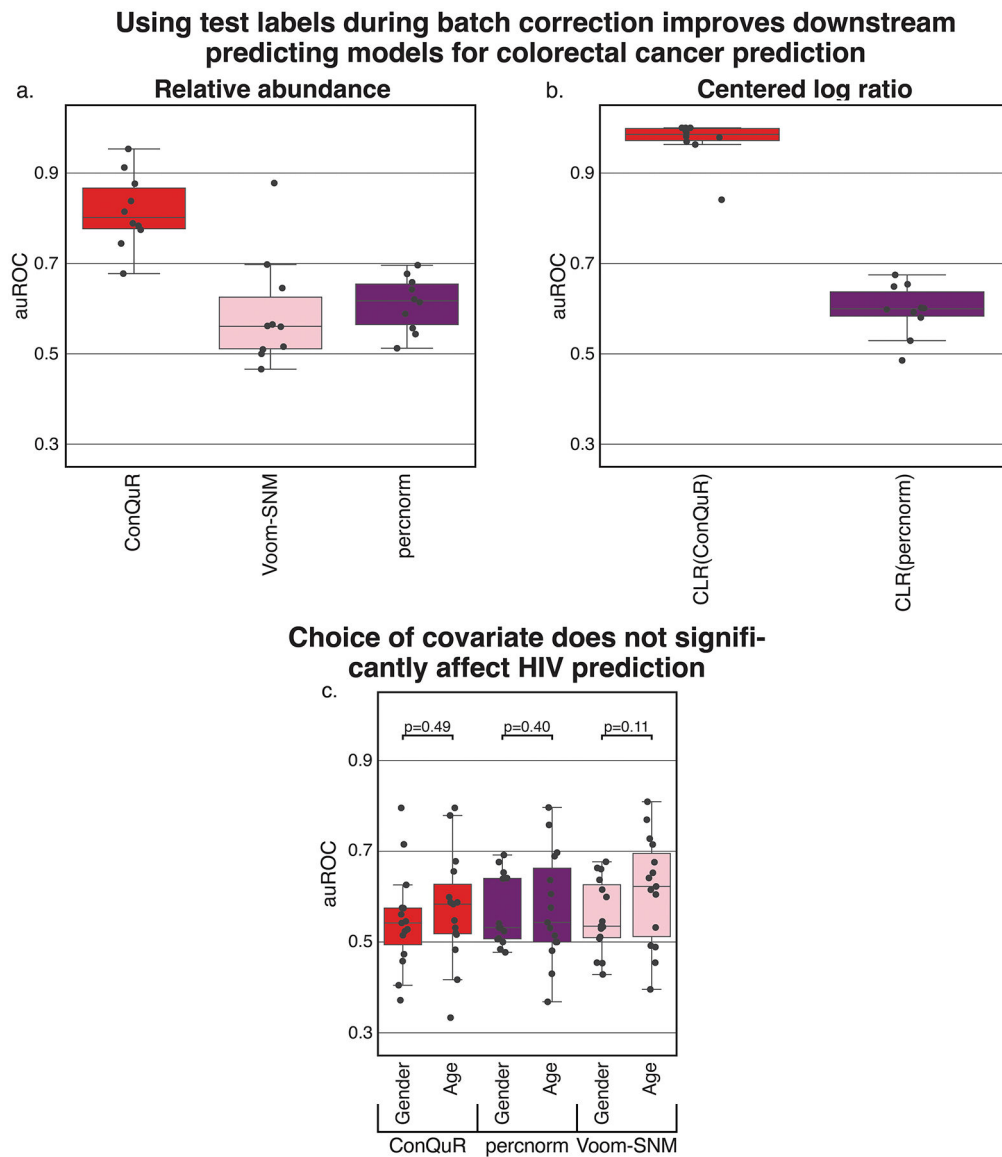


Log-additive DEBIAS-M outperforms alternative methods in benchmarks in centered-log-ratio space



Extended Data Figure 2 | DEBIAS-M maintains its performance in log space and without observing the test set during training.

a, The progression of cross-batch similarity loss as a fitted online DEBIAS-M model adapts to samples from a previously unobserved study, by solely minimizing the cross-batch similarity loss. **b**, the predictive performance of the fitted online DEBIAS-M model throughout the adaptation iterations. Although not directly used during the adaptation itself, the auROC of the model's prediction on the held-out test increases as the cross-batch similarity increases. **c**, Box and swarm plots (Box, IQR; line, median; whiskers, nearest point to 1.5*IQR) comparing the auROC of DEBIAS-M (fitted and evaluated using the strategy in Extended Data Fig. 1a) with "Online DEBIAS-M" (fitted and evaluated using the strategy in Extended Data Fig. 1c) on each held-out study, for the benchmarks used in Fig. 2a–c. Online DEBIAS-M demonstrated equivalent predictive performance on held-out studies. *p* - two-sided Wilcoxon signed-rank test. See Supplementary Tables 1,3,4 for information on studies and sample sizes. **d-g**, Same as Fig. 2, but comparing log-additive DEBIAS-M to batch-correction methods on clr-transformed data. Most methods operate only in relative abundance or count space, and we therefore transformed their outcome with centered log transformation. For a fair comparison, we also clr-transformed the output of regular DEBIAS-M (denoted "CLR(DEBIAS-M)") PLSDA-batch was run on clr-transformed abundances. Voom-SNM is not included as it cannot operate on clr-transformed values and its output is not in non-negative relative abundance or count space. *, Fisher's multiple comparison of one-sided DeLong tests $p < 0.01$ vs. DEBIAS-M. See Supplementary Table 7 for exact *p* values.



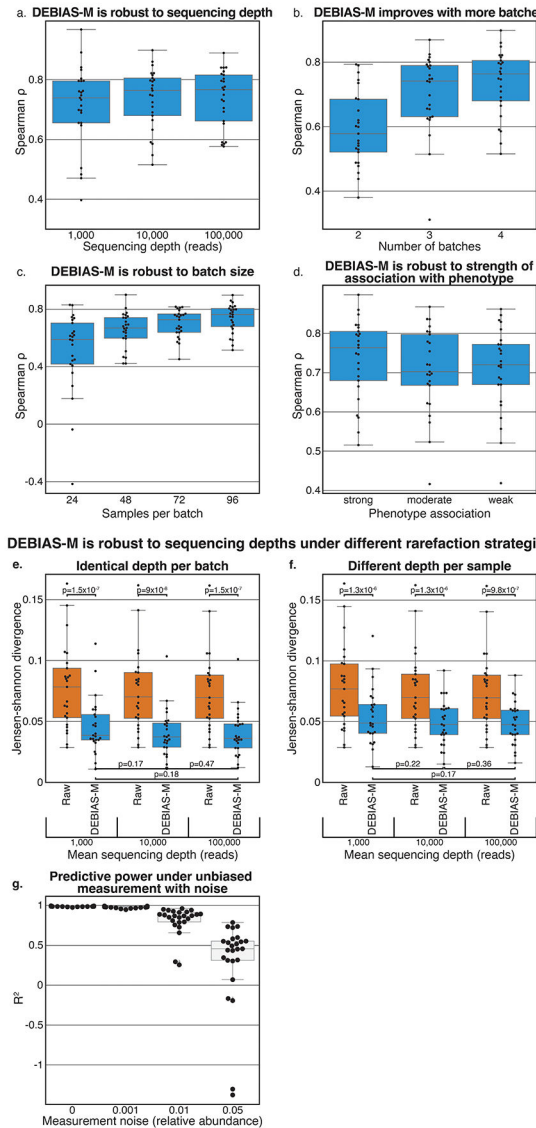
Extended Data Figure 3 |. Using test set labels during batch correction can drastically increase measured predictive performance in downstream benchmarks.

a-b, The same cross-study colorectal cancer prediction benchmark as in Fig. 2b and Extended Data Fig. 2e, respectively, but percnorm, ConQuR, and Voom-SNM were provided all colorectal cancer labels, including for the test set, during batch correction (as in Extended Data Fig. 1d). The prediction accuracy (auROC) of certain methods inflated drastically beyond the results observed in the primary benchmark (Fig. 2b and Extended Data Fig. 2e), highlighting potential issues with assessing a batch-correction method by measuring the ability of a downstream machine learning model to predict information used during batch correction. This trend is consistent in both relative abundance space (**a**) and center log ratio (**b**). Voom-SNM is not run for (**b**) as its output is neither in non-negative relative abundance nor in count space. See Supplementary Table 2 for information on studies and sample sizes.

c, The same cross-study HIV prediction benchmark as in Fig. 2a, comparing the accuracy on each held-out study for ConQuR, percnorm, and Voom-SNM when provided with gender or

age as a covariate (see Extended Data Fig. 1b). See Supplementary Table 1 for information on studies and sample sizes. p , one-sided Wilcoxon signed-rank test. Across all panels: Box, IQR; line, median; whiskers, nearest point to $1.5 \times \text{IQR}$. One point corresponds to the specified model and dataset's predictive performance on a held-out study.

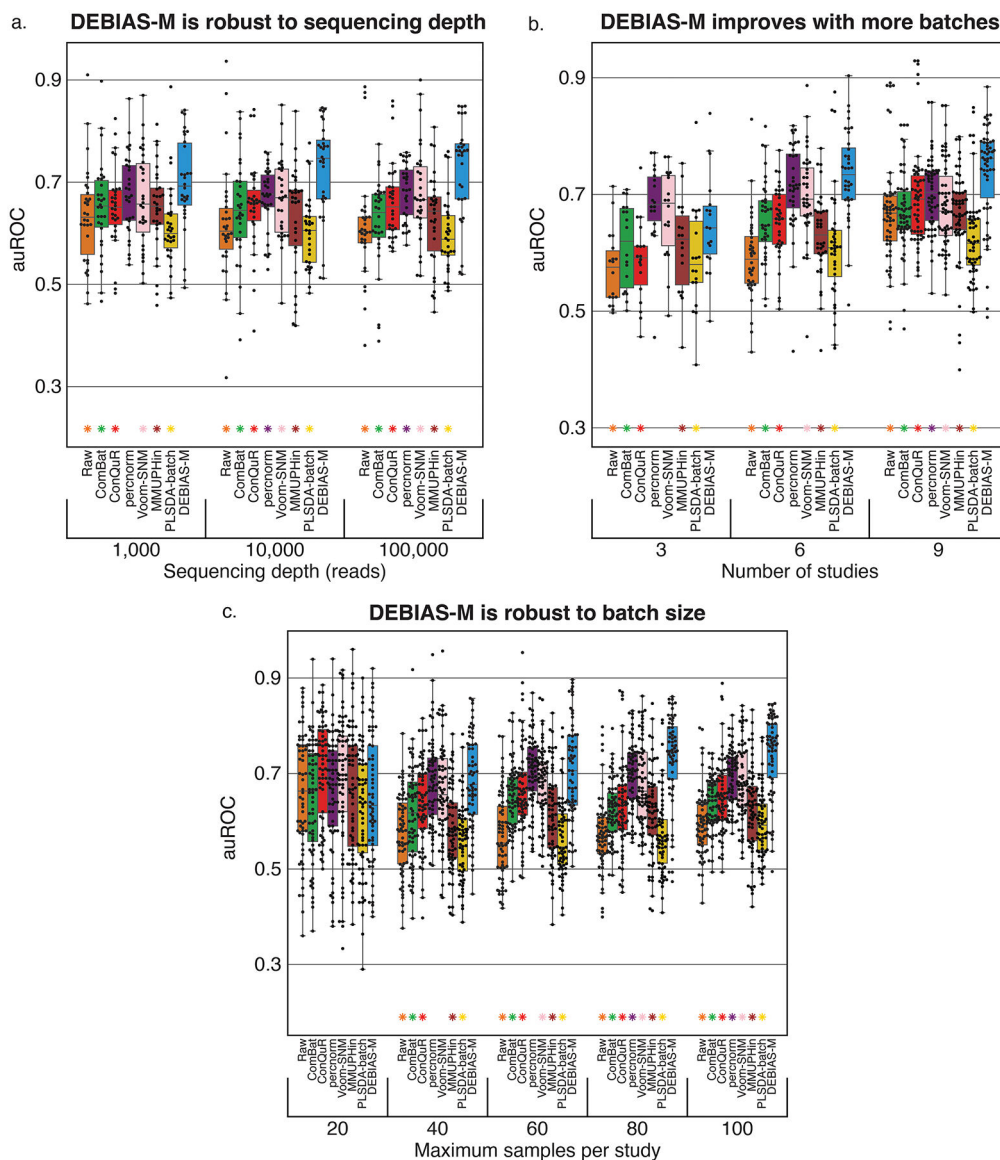
DEBIAS-M accurately infers processing biases in *in silico* simulations under its model



Extended Data Figure 4 | DEBIAS-M is robust to dataset characteristics in simulations under its model.

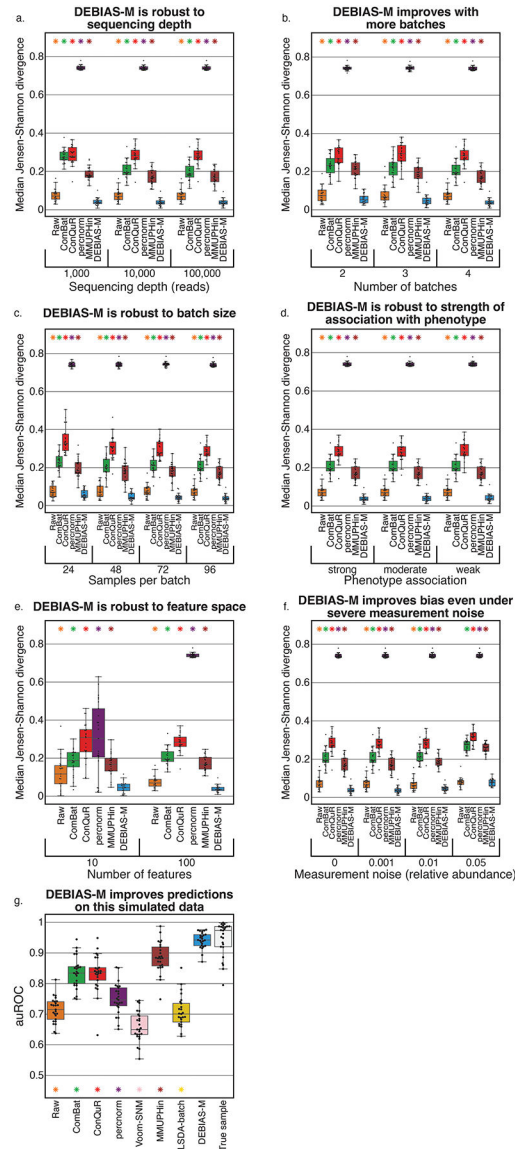
a-d, Box and swarm plots, showing results for the same *in silico* simulations as in Fig. 3a-d, but evaluating the ability of DEBIAS-M to infer the ground-truth biases ($N=25$ experiments per boxplot). DEBIAS-M shows consistent ability across varying sequencing depths (a) and strength of association with phenotype (b). However, it shows improvement with the number of batches (b; especially >2) and batch size (c; especially >40). $N=25$ experiments per box. **e-f**, Box and swarm plots similar to Fig. 3a, except in (e) all samples in each batch have a batch-specific read depth (with a global mean across batches), which

simulates rarefaction to a batch-specific threshold; and in (f) each sample has its own read depth, with a different mean for each batch (and a global mean across batches), simulating non rarefaction. DEBIAS-M is robust to sequencing depth in all cases. *p*: Raw vs. DEBIAS-M one-sided Wilcoxon signed-rank test; DEBIAS-M vs. DEBIAS-M two-sided Mann-Whitney *U* test. **g**, The strength of association between simulated phenotype and phenotype predictions obtained using ground-truth weights and biases on the microbiome data (Methods), demonstrating the effect of increasing measurement noise in Fig. 3f. Across panels: Boxplots: line, median; box, IQR; whiskers, nearest point to 1.5*IQR. One point corresponds to one simulation, evaluating DEBIAS-M's (a-f) or the ground-truth (g) performance on the held-out batch of that simulation.



Extended Data Figure 5 | DEBIAS-M is robust across dataset characteristics using non-simulated data.

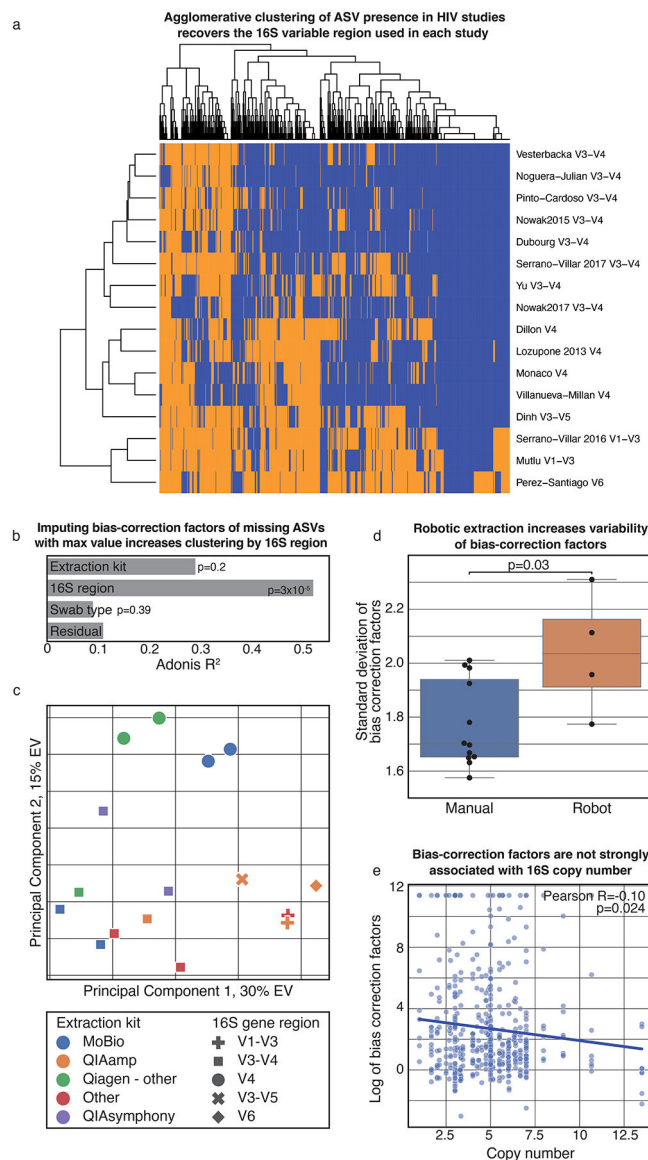
a-c, Box and swarm plots (line, median; box, IQR; whiskers, nearest point to 1.5*IQR) showing results for simulations performed by subsetting the colorectal cancer dataset (analyzed in Fig. 2b), evaluating accuracy of classifying cancer presence using the gut microbiome under varying sequencing depth (a), number of samples per study (b) and the number of studies (c) to each held-out study (see Supplementary Table 3 for information on studies and sample sizes). DEBIAS-M significantly outperforms alternative methods in all benchmarks, but demonstrates reduced performance when the number of samples per study drops to 20. *, one-sided Wilcoxon signed-rank $p < 0.05$ between DEBIAS-M and marked method. See Supplementary Table 7 for exact p values.



Extended Data Figure 6 | DEBIAS-M outperforms alternative methods across dataset characteristics in simulations under its model.

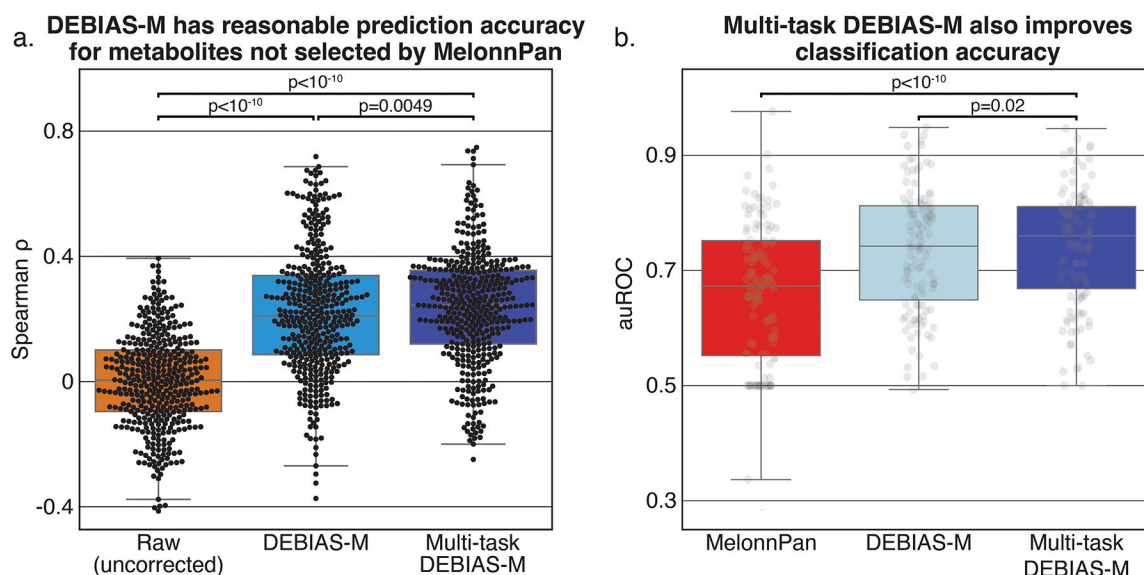
a-f, Similar to Fig. 3a-f, respectively, but showing comparisons to alternative methods that produce outputs in counts or relative abundance space. Each point corresponds to the

held-out batch of a single simulation. N=25 experiments per box. Note that ConQuR and percnorm are receiving the phenotype for the test-set in these analyses (as in Extended Data Fig. 1d). **g**, Box and swarm plots (line, median; box, IQR; whiskers, nearest point to 1.5*IQR) showing the accuracy (as auROC) of predicting the phenotype evaluated on held-out test batches. In this simulation, ConQuR, percnorm, and Voom-SNM receive a simulated covariate (Methods) during batch correction (as in Extended Data Fig. 1b). As simulations are under the generative model of DEBIAS-M, they illustrate how the assumptions of other methods differ from the multiplicative bias model. *, two-sided Wilcoxon signed-rank $p < 0.05$ between DEBIAS-M and marked method. See Supplementary Table 7 for exact p values.



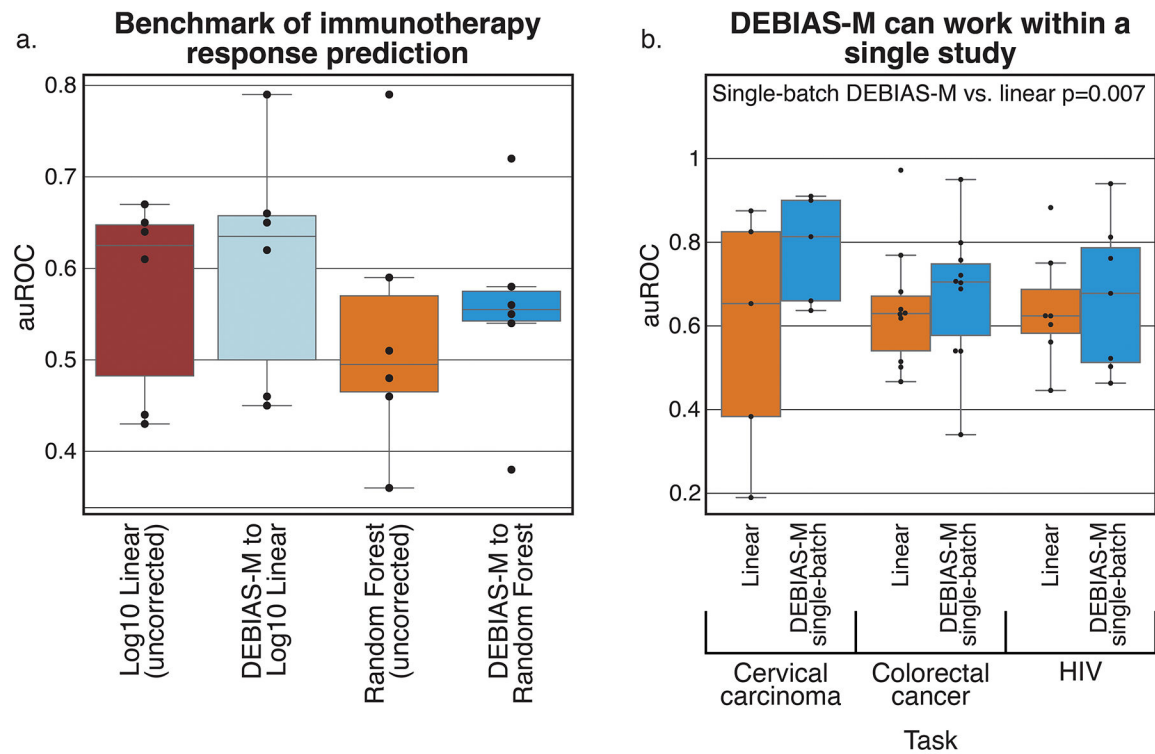
Extended Data Figure 7 | DEBIAS-M inference yields biological insights into sequencing bias.

Analyses of a fitted DEBIAS-M model on the collection of HIV studies used in Figs. 2a, 4, with bias-correction factors for species not found in a certain study imputed to the largest observed factor across all datasets. **a**, Heatmap illustrating the presence (blue) and absence (orange) of each OTU across each of the HIV studies analyzed, displayed using agglomerative clustering (Methods). The OTU detection patterns of the different studies cluster according to the 16S region amplified. **b**, Adonis PERMANOVA explained variance and *p* values for the effect of different experimental factors (Supplementary Table 1) on these bias correction factors. **c**, PCA plot of the bias-correction factors inferred by DEBIAS-M, same as Fig. 4c, but with bias-correction factors for OTUs not found in a certain study imputed to the largest observed factor across all datasets. Color represents extraction kit type and shape the 16S rRNA region used. **d**, Box and swarm plots (Box, IQR; line, median; whiskers, nearest point to 1.5*IQR) showing the standard deviation of bias-correction factors for each of 17 studies (Supplementary Table 1), comparing those with manual and robotic processing. *p*, one-sided Mann-Whitney *U* test. **e**, Scatterplot showing the bias-correction factors inferred by DEBIAS-M plotted versus the 16S copy number of the same species.



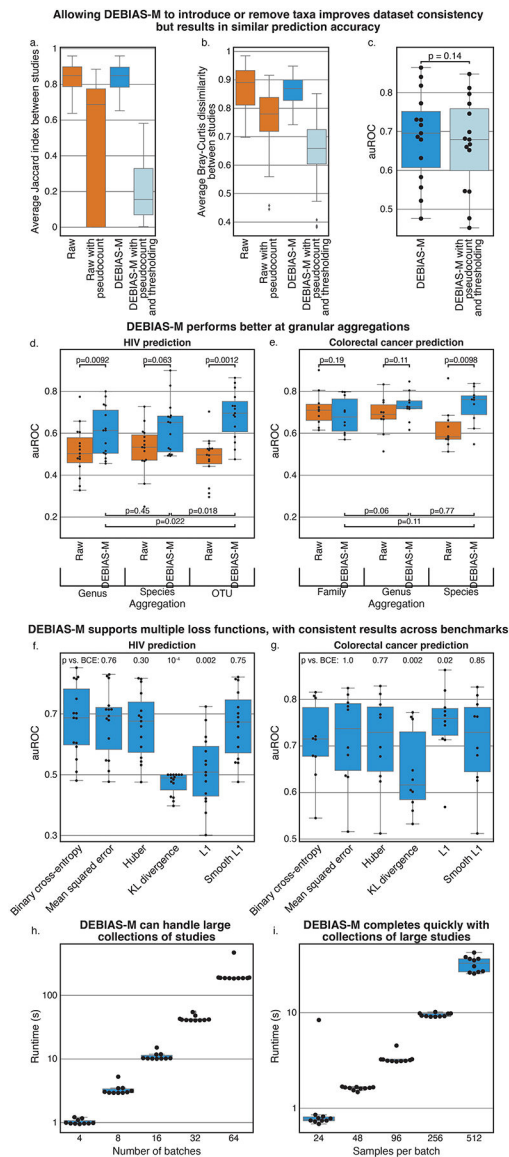
Extended Data Figure 8 | Multi-task DEBIAS-M improves classification and regression of metabolite levels.

a, Same as Fig. 5c, showing 399 metabolites with >50 observations for which MelonnPan⁵⁰ does not provide a prediction, using samples from 218 individuals. **b**, Same as (a), but classifying samples where a metabolite had levels that were higher than the median, for all metabolites. This analysis was performed for 120 metabolites selected by MelonnPan out of 509 metabolites that had values lesser and greater than their median in both batches, such that a classification task could be used to train and evaluate a model. Samples from 218 individuals were used. The multi-task version of DEBIAS-M outperformed both its single-task version (one-sided Wilcoxon signed-rank $p = 0.02$, as well as MelonnPan ($p = 5.6 \times 10^{-14}$)). Box, IQR; line, median; whiskers, nearest point to 1.5*IQR.



Extended Data Figure 9 | DEBIAS-M improves cross-study prediction of melanoma immunotherapy response and works within a single study.

a, Box and swarm plots of auROCs, each evaluating the cross-study generalization performance models using gut microbiome data to predict immunotherapy response in melanoma patients (defined as 12-month progression-free survival⁵¹). Each auROC is calculated on a held-out study. See Supplementary Table 6 for information on studies and sample sizes. ‘Log10 Linear’ denotes the pipeline used by Lee et al.⁵¹. Preprocessing with DEBIAS-M shows a consistent albeit small improvement across all studies, with a particularly strong effect for one study. **b**, Box and swarm plots of auROCs, each representing the aggregate accuracy over the held-out points for a logistic regression model evaluated under 5-fold cross validation. ‘Cervical carcinoma’, ‘Colorectal cancer’ and ‘HIV’ correspond to the benchmark datasets from Fig. 2d, Fig. 2b, and Fig. 2a, respectively, with information on studies and samples sizes in Supplementary Tables 5, 2 and 1. *p*, one-sided Wilcoxon signed-rank test comparing between linear and DEBIAS-M single-batch across both datasets. Box, IQR; line, median; whiskers, nearest point to 1.5*IQR.



Extended Data Figure 10 |. Methodological consideration for DEBIAS-M.

a-c, An analysis performed on the HIV benchmark (analyzed in Fig. 2a) using a standard implementation of DEBIAS-M as well as one using data with an added pseudocount of one, which allows DEBIAS-M to add non-measured taxa, and in which feature values with relative abundance below a threshold of 10^{-4} were replaced with zero, which allows DEBIAS-M to completely remove a taxon. The average pairwise Jaccard index (a) and Bray-Curtis dissimilarity (b) between studies are lower for DEBIAS-M with pseudocount and thresholding, indicating an improved ability to reduce batch effects. However, this does not result in improved accuracy in the benchmark itself (c). **d,e**, Box and swarm plot of auROCs, each evaluating the generalization performance of DEBIAS-M to a held out study from the HIV (Fig. 2a; d) and colorectal cancer (Fig. 2b; e) benchmarks. In each case, we have applied DEBIAS-M to data aggregated to different taxonomic levels, as available from each of the datasets. DEBIAS-M becomes less effective with more aggregation,

and, for colorectal cancer prediction, actually reduces predictive performance compared to uncorrected data after aggregation to family level. **f,g**, Box and swarm plot of auROCs, each evaluating the generalization performance of DEBIAS-M to a held out study from the HIV (Fig. 2a; f) and colorectal cancer (Fig. 2b; g) benchmarks using different prediction loss functions. Most alternative loss functions are not significantly different from the binary cross-entropy used as default by DEBIAS-M for classification. The L^1 loss shows significant improvement on colorectal cancer prediction ($p=0.02$; e), but a significant reduction in performance on the HIV benchmark ($p=0.002$; d). **h,i**, The total runtime of DEBIAS-M on a standard laptop for simulated datasets with an increasing number of batches, each containing 96 samples (h), as well as for 8 batches, each with an increasing number of samples (i). DEBIAS-M runs in <15 minutes (median 187 seconds) even for 64 batches with 6,144 samples in total, and in <43 seconds (median 33 seconds) for 8 batches with 4,096 samples in total. Box, IQR; line, median; whiskers, nearest point to $1.5 \times \text{IQR}$; p, two-sided Wilcoxon signed-rank test.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability

All datasets analyzed in this study are publicly available. The HIV dataset is available from Synapse (<https://www.synapse.org/#!Synapse:syn18406854>). The colorectal cancer and melanoma immunotherapy datasets are available through the R curatedMetagenomicData package⁶⁰. The cervical neoplasia datasets were compiled from data provided with each publication, with information detailed in Supplementary Tables 4,5. The paired vaginal microbiome and metabolite data were obtained from Supplementary Data 2 of ref.⁴⁸ (also available from dbGaP, accession phs001739.v1.p1) and Supplementary Table 1 of ref.⁴⁷ (also available from MetaboLights, accession number MTBLS702). Available accession codes for all other datasets are listed in Supplementary Tables 1,3–6.

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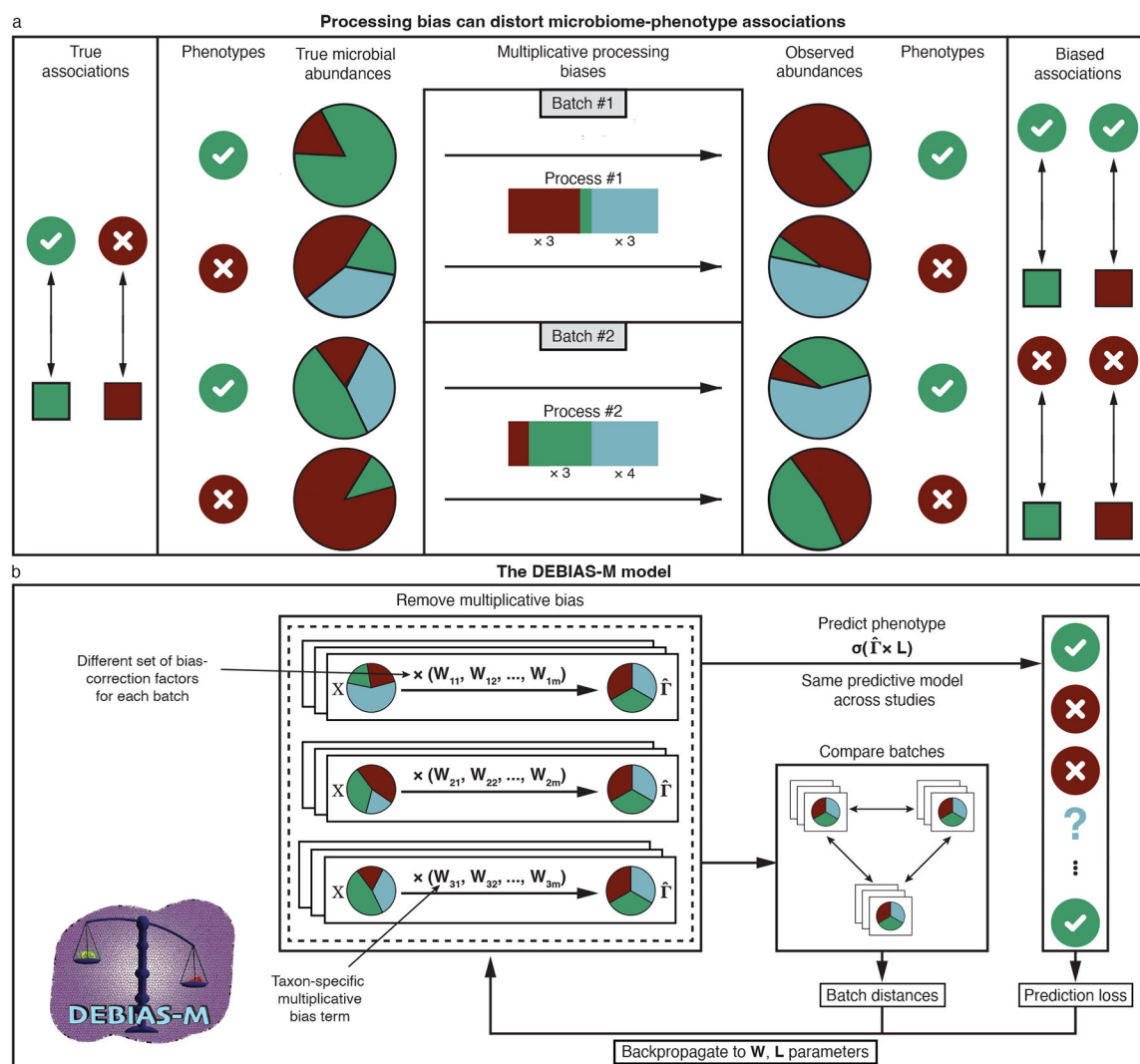


Figure 1 |. Processing-bias correction with DEBIAS-M.

a, An illustration of our assumptions underlying the data-generating process: The “real” relative abundances of a given sample are distorted by batch-specific per-taxa multiplicative biases, which can both obfuscate the true signals or generate artifactual ones, particularly in cross-batch analyses. **b**, DEBIAS-M iteratively learns per-species multiplicative biases for each batch which minimize the distances between the batches and maximize linear associations to a phenotype of interest. The combination of both prediction loss and batch distances prevents DEBIAS-M from removing meaningful variation. Samples without available phenotype (missing or hidden) are used only when calculating the batch-wise differences (or not at all for online learning; Methods).

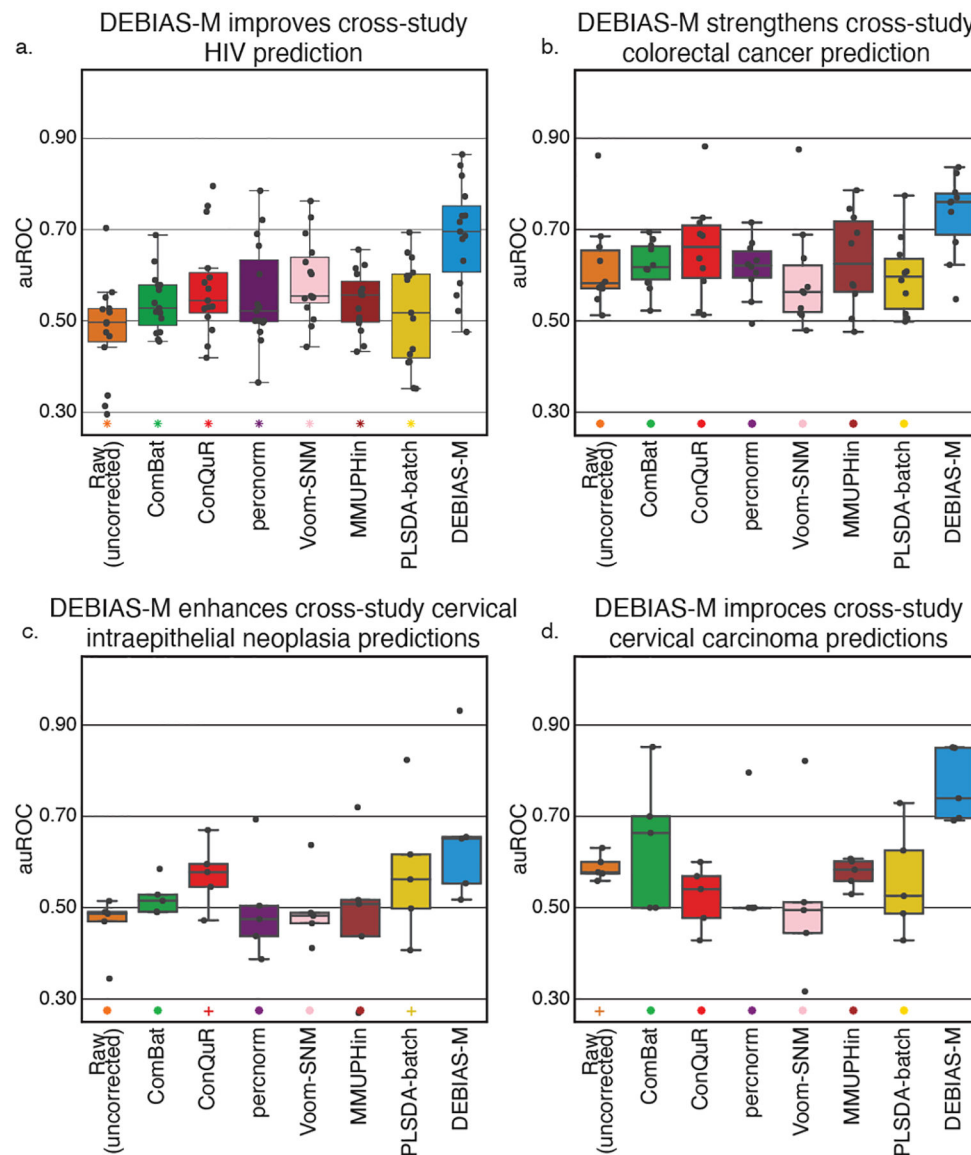


Figure 2 | DEBIAS-M outperforms batch-correction methods in cross-study prediction benchmarks.

a-d, Box and swarm plots (Box, IQR; line, median; whiskers, nearest point to 1.5*IQR) of auROCs, each evaluating the generalization performance of logistic regression models to a held-out study, predicting: HIV from gut microbiome data (**a**; N=13–233 samples per study; Supplementary Table 1; Methods); colorectal cancer from gut microbiome data (**b**; N=53–128 samples per study; Supplementary Table 3); cervical intraepithelial neoplasia from cervicovaginal microbiome data (**c**; N=29–82 samples per study; Supplementary Table 4); and cervical carcinoma from cervicovaginal microbiome data (N=26–77 samples per study; Supplementary Table 5). * and + denotes $p < 0.01$ and $p < 0.05$, respectively, for DEBIAS-M vs marked method, via Fisher's multiple comparison of one-sided DeLong tests. See Supplementary Table 7 for exact p values.

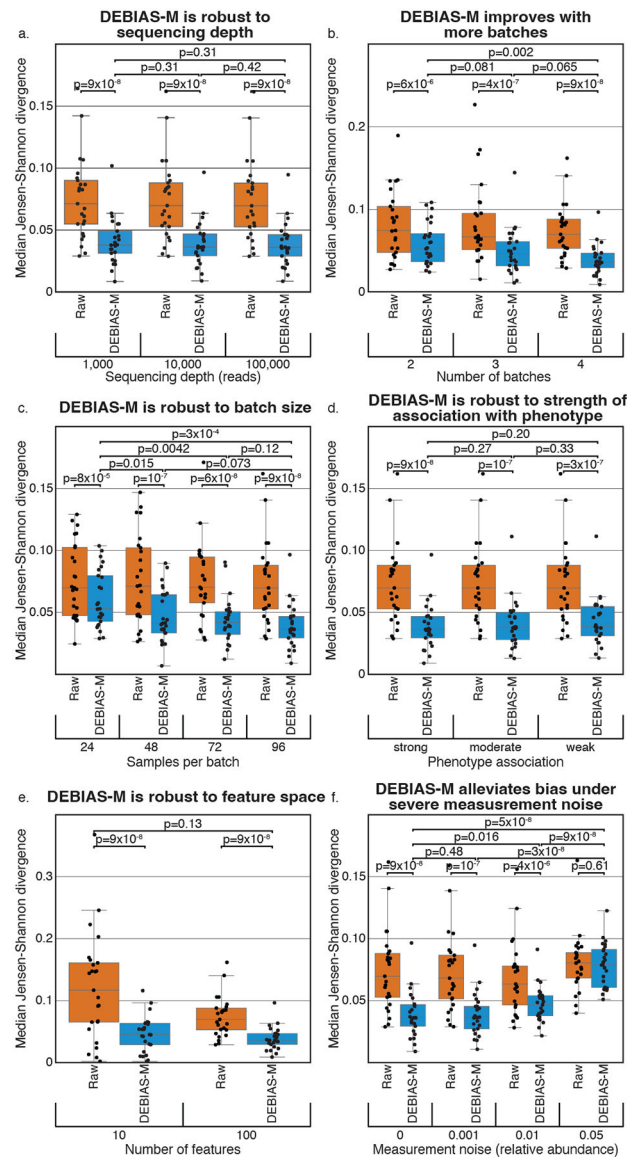


Figure 3 | DEBIAS-M is robust across dataset characteristics in simulations.

Box and swarm plots (line, median; box, IQR; whiskers, nearest point to $1.5 \times \text{IQR}$), showing results for *in silico* simulations built under our generative model, with additional noise in some simulations. DEBIAS-M consistently brings the relative abundance of samples closer to the known ground truths under varying sequencing depth (a), number of batches (b), batch size (c), strength of association with phenotype (d), number of microbiome features (e), and measurement noise added in addition to bias (and independent of it; f; see Extended Data Fig. 4g for the impact of this noise on phenotypic associations). N=25 experiments per box. In all cases except very high measurement noise, DEBIAS-M improves the representation of the simulated microbiomes. *p*: Raw vs. DEBIAS-M one-sided Wilcoxon signed-rank test; DEBIAS-M vs. DEBIAS-M two-sided Mann-Whitney *U* test, due to an unequal number of possible simulations.

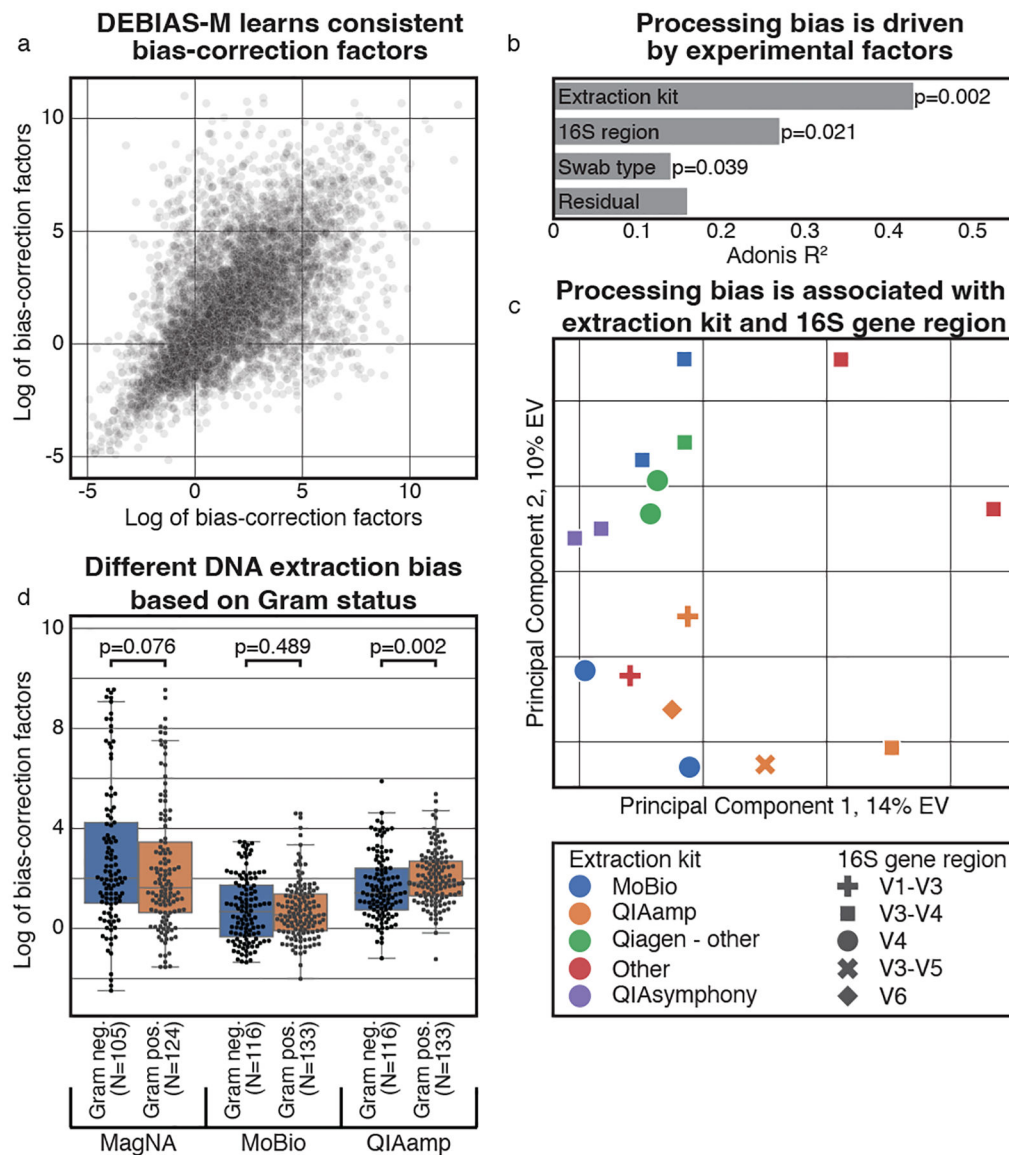


Figure 4 | DEBIAS-M infers biases that are stable, interpretable, and consistent with experimental processes.

a, Scatterplot of bias-correction factors inferred by DEBIAS-M separately for two halves of the same held-out study, evaluated across all studies from the HIV benchmark. Inferred biases are consistent for the same batch (Pearson $R=0.59$; $p<10^{-10}$). **b**, Adonis PERMANOVA explained variance and p-values for the association of different experimental factors (Supplementary Table 1) with the bias-correction factors inferred by DEBIAS-M for each HIV study. Extraction kit type has the strongest effect on bias. **c**, PCA plot of the bias-correction factors inferred by DEBIAS-M. Color represents the extraction kit type and shape the 16S rRNA region used. **d**, Box and swarm plots (line, median; box, IQR; whiskers, nearest point to $1.5 \times \text{IQR}$) showing the bias-correction factors inferred for each taxon by DEBIAS-M (y-axis) for taxa from the HIV benchmark, stratified by sequencing kit and Gram status. p , two-sided Mann-Whitney U test.

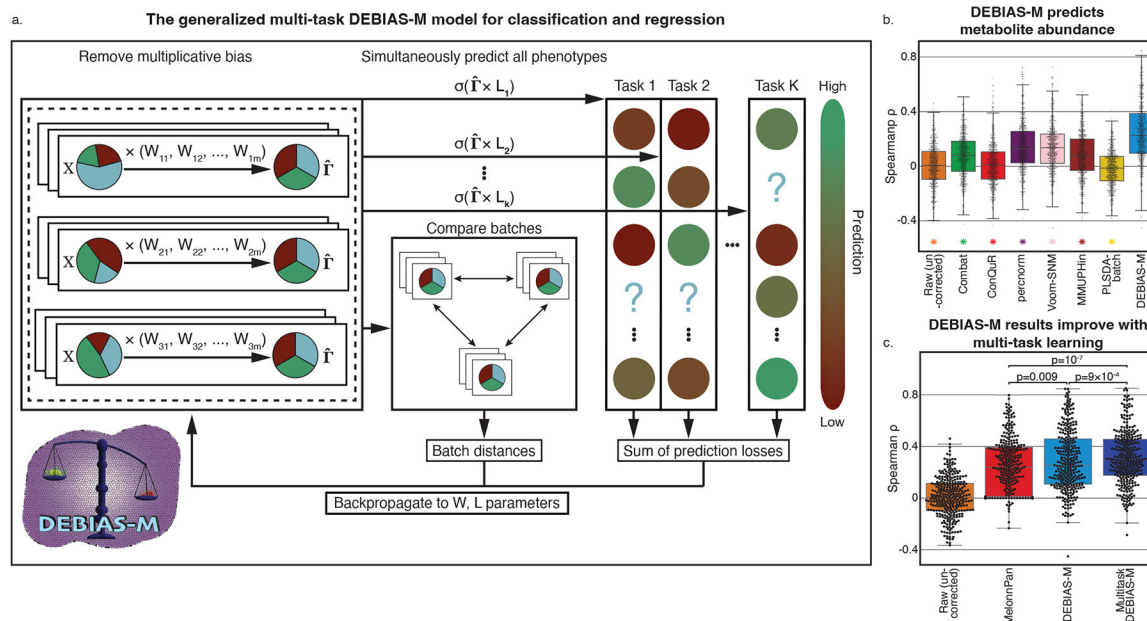


Figure 5 |. Regression and multi-task learning with DEBIAS-M improves metabolite predictions.

a, Description of how DEBIAS-M could be extended for both regression and for multi-task learning. In multi-task DEBIAS-M, a single set of bias-correction factors is learned per study while multiple prediction tasks are considered separately (Methods). **b**, Box and swarm plots of correlation coefficients (Spearman) from regression models predicting the levels of each of 684 metabolites separately using samples from 218 individuals, evaluated for cross-batch generalization. *, one-sided Wilcoxon signed-rank $p < 10^{-10}$ vs. DEBIAS-M. See Supplementary Table 7 for exact p values. **c**, Box and swarm plots, similar to (b), but showing predictions made by both MelonnPan⁵⁰ and multi-task DEBIAS-M. Shown are results for predictions of each of 297 metabolites selected by MelonnPan⁵⁰. The multitask version of DEBIAS-M outperformed both its single-task version ($p = 9 \times 10^{-4}$) and MelonnPan ($p = 10^{-7}$). See Extended Data Fig. 8 for performance on metabolites not selected by MelonnPan (Extended Data Fig. 8a) as well as for a benchmark of multi-task DEBIAS-M on a classification task (Extended Data Fig. 8b). p , one-sided Wilcoxon signed-rank test. Box, IQR; line, median; whiskers, nearest point to $1.5 \times \text{IQR}$.

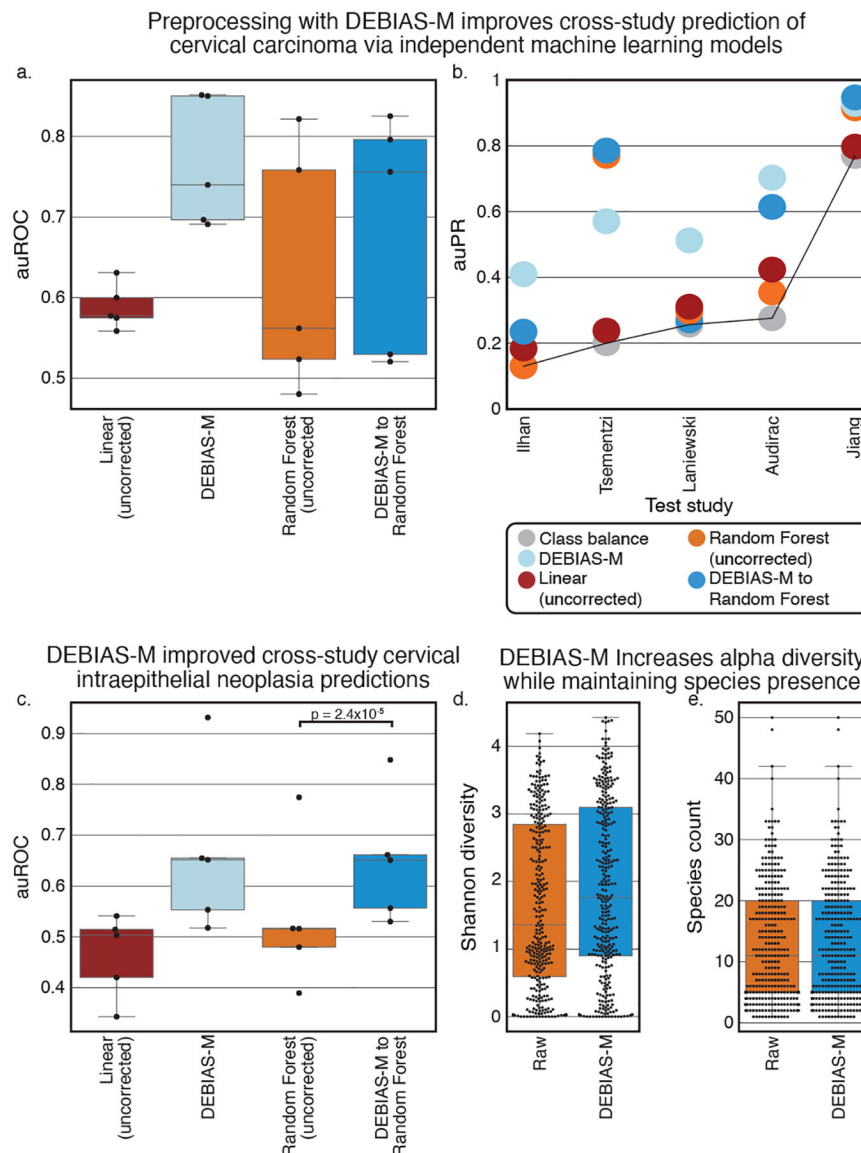


Figure 6 | DEBIAS-M improves cross-study prediction of cervical cancer.

a,b, Plots showing the auROCs (a) and auPRs (b), each evaluating the cross-study generalization performance of linear and random forest models predicting cervical carcinoma from cervicovaginal microbiome data, evaluating DEBIAS-M as a pre-processing step (as in Extended Data Fig. 1e; N=26–77 per study; Supplementary Table 5). **c,** Same as (a) for prediction of cervical intraepithelial neoplasia from cervicovaginal microbiome data (N=26–83 samples per study; Supplementary Table 4). *p*, Fisher's multiple comparison of one-sided DeLong tests comparing DEBIAS-M vs. the marked method. **d,** Shannon diversities of both the raw relative abundance of 323 cervicovaginal microbiome samples and of the same samples after processing with DEBIAS-M, across all studies. DEBIAS-M increases the diversity of these samples (two-sided Wilcoxon signed-rank $p < 10^{-10}$). **e,** Count

of species presence for the same samples. By design, DEBIAS-M does not change the number of observed species. Box, IQR; line, median; whiskers, nearest point to $1.5 \times \text{IQR}$.